

Module manual for Civil Engineering (Bachelor (1- Subject))



Examination Regulation Field



Module offer



Examination offer



Teaching offer

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Examination Regulation Title & Version:
Civil Engineering (SPO Version / 2020)

Title	Civil Engineering
Short title	BSBau
Version	2020
Study/Qualification Objectives	The graduates of this bachelor's degree program have become familiar with using a holistic system approach. They have the necessary basic mathematical and scientific knowledge, technical expertise in the discipline of civil engineering as well as the knowledge of the methods commonly used in this field. They have developed an awareness of the interdisciplinary dimensions and implications of their work and are able to have a reflected view on their responsibility to society. Students who have earned a bachelor's degree have the following qualifications: • They can analyze complex problems from the specialized professional fields of civil engineering. They are able to prepare these problems in engineering terms in order to develop and evaluate innovative solution concepts. • They master the scientific methods of analyzing problems in their basic structure and have introductory knowledge of theoretical problem description and mathematical modeling in this field. • They can recognize connections between their discipline and related fields and discuss them with representatives of these disciplines. • They are empowered to expand their theoretical knowledge and are able to bring it to bear and apply it to future technologies. • In particular, they are able to apply the methods of their chosen specializations to identify, analyze, and solve challenging problems in their discipline. • They can successfully work on synthesis problems, especially in the context of complex systems, with a balanced consideration of technical, scientific, and social constraints. • They have become acquainted with exemplary selected fields of technology and have built bridges between engineering fundamentals and applications related to a professional field. • They have acquired many different methods and non-technical skills that enable them to work in a broad professional field.
Qualification Profile	
Additional information	

Module titel	Applied Statistics (Compulsory subject)
Identifier	3010717
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2007
Valid until	-
Module level	Bachelor
Content	Matrix algebra and solving systems of linear equations; terms of descriptive and inductive statistics (location and dispersion parameters); random variables and probability distributions; covariance matrix of linear transformations (variance / covariance - propagation); linearization of non-linear transformations; regression and correlation analysis; methods of parameter estimation; confidence intervals and hypothesis testing
Learning Objectives/ Learning Outcomes	Understanding the formulation and modeling of engineering processes in systems of linear equations; reliable assessment of precision and quality in construction processes; ability to calculate the stepwise propagation of accuracies (variances/covariances) in production processes; significant judgement of series of measurements (sampling and testing procedures, outlier search); familiarity with the formulation and estimation of functional dependencies
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Übungsumdrucke; Lehrbuch W. Benning: Statistik in Geodäsie, Geoinformation und Bauwesen. Wichmann Verlag; Lehrbuch Witte/Schmidt: Vermessungskunde und Grundlagen der Statistik für das Bauwesen. Wichmann Verlag
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the exam.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Universitätsprofessor Dr.-Ing. Jörg Blankenbach
ECTS Credits	3
Contact time (WSH)	3
Examination duration (min)	120
Total hours (h)	90,0
Contact hours (h)	45,0
Self-study hours (h)	45,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam Applied Statistics (301071701)	1st semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Tutorial Applied Statistics	1st semester	no semester recommended	-	3

Module titel	Hydromechanics (Compulsory subject)
Identifier	3020962
Version	V1
Duration (Semester)	two semesters
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2019
Valid until	-
Module level	Bachelor
Content	Hydromechanik I: Physikalische Eigenschaften der Flüssigkeiten; Mathematische Beschreibung der Bewegung von Flüssigkeiten; Hydrostatik und Hydrodynamik; Impulssatz; Rohrströmung; Turbulenz; Hydromechanik II: Laminare und turbulente Rohrströmung; Gerinneströmung; Grundwasserströmung; Überströmung von Wehren
Learning Objectives/ Learning Outcomes	Hydromechanik I: Die Studierenden erlernen über ein profundes Verständnis der Grundlagen der Hydromechanik ein Rüstzeug für die eigenständige Bemessung hydrostatisch und hydrodynamisch belasteter Bauteile. Ziel ist das Verständnis der thematischen Breite vor der Abbildung der vollständigen theoretischen Tiefe. Dabei entwickeln die Studierenden Lernstrategien zur Aneignung neuer, im schulischen Bereich nicht behandelter und komplexer Theorien. Aufgrund der Komplexität der behandelten Themen werden die Studierenden die Fähigkeit zur Bildung von Analogien zu anderen physikalischen Disziplinen (bspw. Aerodynamik) lernen. Hierdurch wird das Verständnis von Alltagsphänomenen erleichtert. Hydromechanik II: Die Studierenden vertiefen ihre bestehenden Kenntnisse (Hydromechanik I) in Richtung eines profunden Verständnisses hydraulischer Phänomene. Hierdurch wird die Befähigung zur Übertragung theoretischer Materie in die wasserbauliche Praxis weiter gefördert. Die Studierenden werden theoretische Probleme selbständig in anschauliche Teilaspekte gliedern und lösen.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdruck Hydromechanik
Language	German
Examination Terms	Graded written exam. Prerequisite for the participation in the exam: passed term papers on Hydromechanics I and II.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Univ.-Prof. Dr.-Ing. Holger Schüttrumpf
ECTS Credits	6
Contact time (WSH)	4

Examination duration (min)	-
Total hours (h)	180,0
Contact hours (h)	60,0
Self-study hours (h)	120,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term Paper Hydromechanics I (302096202)	3rd semester	no semester recommended	0	0
Term Paper Hydromechanics II (302096203)	4th semester	no semester recommended	0	0
Exam Hydromechanics (302096201)	4th semester	no semester recommended	6	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and exercise Hydromechanics I	3rd semester	no semester recommended	-	2
Lecture and exercise Hydromechanics II	4th semester	no semester recommended	-	2

Module titel	Mathematics I (Compulsory subject)
Identifier	1115630
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2017
Valid until	-
Module level	Bachelor
Content	Basic knowledge on real and complex numbers, convergence of sequences and series Analysis of functions in one variable, limits, continuous functions; differential calculus, integration
Learning Objectives/ Learning Outcomes	Understanding of mathematical notions, ideas and methods Knowledge of important functions Development of solution strategies for mathematical Problems Application of mathematical models to specific problems
(Study-Specific) Prerequisites	Zulassungsvoraussetzung zur Teilnahme an der Lehrveranstaltung: keine; Zulassungsvoraussetzung zur Teilnahme an der Klausurarbeit: aktive Teilnahme an Übungen und ggf. angebotenen Wissensstandskontrollen (detaillierte Regelung wird jeweils vor Semesterbeginn bekanntgegeben). Erwartete Vorkenntnisse: Schulmathematik
(recommended) Requirements	Mathematics of high school level
References	Meyberg, Vachenauer: Höhere Mathematik, Springer-Verlag Burg, Haf, Wille: Höhere Mathematik für Ingenieure, Teubner-Verlag Vetters: Formeln und Fakten im Grundkurs Mathematik, Teubner-Verlag ggf. Skriptum zur Mathematik I
Language	German
Examination Terms	Admission requirements for participation in the written exam: active participation in the exercises and testing knowledge (detailed regulation will be announced before the semester begins) Written exam (150 min), Grading: graded, Weight: 100 %. By means of voluntary controls of the level of knowledge, points can be collected. For the first examination attempt in the first semester, these points can be taken into account up to 10 %. Recognition of these points for the second or third examination attempt is not possible. The detailed regulations for bonus points are published in the CMS at the semester start.
Miscellaneous	-
Module coordinator	Modulangebotsorganisator: Modellierungsteamverantwortlicher: Dr. rer. nat. Katja Petzoldt Modulverantwortlicher: Universitätsprofessor Dr. rer. nat. Michael Herty
ECTS Credits	8
Contact time (WSH)	6
Examination duration (min)	150

– Basics in Mathematics and Science
+ Mathematics I (1115630)

Total hours (h)	240,0
Contact hours (h)	90,0
Self-study hours (h)	150,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Tutorial Mathematics I (111563002)	1st semester	no semester recommended	0	0
Exercise Mathematics I (111563003)	1st semester	no semester recommended	0	3
Written exam Mathematics I (111563001)	1st semester	no semester recommended	8	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture Mathematics I	1st semester	no semester recommended	-	3

Module titel	Mathematics II (Compulsory subject)
Identifier	1115631
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2018
Valid until	-
Module level	Bachelor
Content	- linear algebra: linear equations, matrices, determinants, eigenvalues, eigenvectors - differential equations: Bernoulli and separable differential equations, linear equations of high order, systems of linear equations
Learning Objectives/ Learning Outcomes	- Understanding of mathematical notions, ideas and methods - Knowledge of important functions - Development of solution strategies for mathematical problems - Application of mathematical models to specific problems
(Study-Specific) Prerequisites	Zulassungsvoraussetzung zur Teilnahme an der Lehrveranstaltung: keine; Zulassungsvoraussetzung zur Teilnahme an der Klausurarbeit: aktive Teilnahme an Übungen und ggf. angebotenen Wissensstandskontrollen (detaillierte Regelung wird jeweils vor Semesterbeginn bekanntgegeben)
(recommended) Requirements	None
References	Meyberg, Vachenauer: Höhere Mathematik, Springer-Verlag; Burg, Haf, Wille: Höhere Math. für Ing., Teubner-Verlag; Vettters: Formeln und Fakten im Grundkurs Math., Teubner-Verlag; ggf. Skriptum zur Mathematik II
Language	German
Examination Terms	Admission requirements for attending the course: none; Admission requirements for participation in the written exam: active participation in the exercises/additional exercises and testing knowledge Written exam (150 min), grading: graded, weighting: 100 %. By means of voluntary controls of the level of knowledge, points can be collected. For the first examination attempt in the second semester, these points can be taken into account up to 10 %. Recognition of these points for the second or third examination attempt is not possible. The detailed regulations for bonus points are published in the CMS at the semester start.
Miscellaneous	-
Module coordinator	Modulangebotsorganisator: Modellierungsteamverantwortlicher: Dr. rer. nat. Katja Petzoldt Modulverantwortlicher: Universitätsprofessor Dr. rer. nat. Michael Herty
ECTS Credits	8
Contact time (WSH)	6

– Basics in Mathematics and Science
+ Mathematics II (1115631)

Examination duration (min)	0
Total hours (h)	240,0
Contact hours (h)	90,0
Self-study hours (h)	150,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exercise Mathematics II (111563103)	2nd semester	no semester recommended	0	2
Tutorial Mathematics II (111563102)	2nd semester	no semester recommended	0	0
Written exam Mathematics II (111563101)	2nd semester	no semester recommended	8	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture Mathematics II	2nd semester	no semester recommended	-	4

Module titel	Mechanics I (Compulsory subject)
Identifier	3015542
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2017
Valid until	-
Module level	Bachelor
Content	Einführung in die Vektorrechnung; Ebene und räumliche Kräftesysteme (Reduktion, Zerlegung und Gleichgewicht); Schwerpunktberechnung; Auflagerreaktionen und Schnittprinzip; Statische und kinematische Bestimmtheit; Schnittgrößen ebener und räumlicher Stabwerke; Fachwerke; Reibung; Prinzip der virtuellen Verrückung
Learning Objectives/ Learning Outcomes	Sicherer Umgang mit vektoriellen Größen (Zerlegung einer Kraft, Reduktion eines Kräftesystems); Aufstellen und Auswerten von Gleichgewichtsbedingungen; Schwerpunktberechnung; Sicherheit im Erkennen der kinematischen und statischen Bestimmtheit einfacher Stabtragwerke; Sicherheit in der Ermittlung von Auflagerreaktionen und Schnittgrößen ebener und räumlicher Stabtragwerke/Fachwerke
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdruck Mechanik I
Language	German
Examination Terms	Graded written exam. Prerequisite for participation in the written exam: 3 out of 5 certificates of achievement must be passed. If five certificates are passed, the grade of the passed exam will be improved by one grade level. The exact criteria for the acquisition of bonus points are specified in the CMS.
Miscellaneous	-
Module coordinator	Modulverantwortliche: Universitätsprofessorin Dr.-Ing. Stefanie Reese
ECTS Credits	8
Contact time (WSH)	7
Examination duration (min)	0
Total hours (h)	240,0
Contact hours (h)	105,0
Self-study hours (h)	135,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Certificate of performance Mechanics I (301554202)	1st semester	no semester recommended	0	0
Written exam Mechanics I (301554201)	1st semester	no semester recommended	8	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exercise: Mechanics I	1st semester	no semester recommended	-	4
Lecture: Mechanics I	1st semester	no semester recommended	-	3
Tutorial Mechanics I	1st semester	no semester recommended	-	0

Module titel	Mechanics II (Compulsory subject)
Identifier	3015543
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2018
Valid until	-
Module level	Bachelor
Content	Elemente der Elasto-Statik; Allgemeine Beschreibung des Spannungs- und des Verzerrungszustands; Materialgesetz für isotrope, linearelastische Körper; Vollständiges Gleichungssystem der Elasto-Statik; Biegung mit Normal- und Querkraft; Torsion; Differentialgleichung der Biegelinie; Statisch unbestimmte Systeme; Arbeitssätze; Stabilitätsprobleme in der Stabstatik
Learning Objectives/ Learning Outcomes	Sichere Kenntnisse in der Berechnung von Flächenwerten; Sicherheit in der Berechnung von Normalspannungen infolge Biegung; Sicherheit in der Berechnung von Schubspannungen infolge Querkraft und Torsion; Kenntnisse in der Berechnung von Formänderungen (Biegelinie, Arbeitssätze); Fähigkeit zur Berechnung von Verzweigungslasten/Kenntnisse in der Stabilitätstheorie
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdruck Mechanik II
Language	German
Examination Terms	Graded written exam. Prerequisite for participation in the written exam: 3 out of 5 certificates of achievement must be passed. If five certificates are passed, the grade of the passed exam will be improved by one grade level. The exact criteria for the acquisition of bonus points are specified in the CMS.
Miscellaneous	-
Module coordinator	Modulverantwortliche: Universitätsprofessorin Dr.-Ing. Stefanie Reese
ECTS Credits	9
Contact time (WSH)	7
Examination duration (min)	90
Total hours (h)	270,0
Contact hours (h)	105,0
Self-study hours (h)	165,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Certificate of performance Mechanics II (301554302)	2nd semester	no semester recommended	0	0
Written exam Mechanics II (301554301)	2nd semester	no semester recommended	9	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture: Mechanics II	2nd semester	no semester recommended	-	3
Tutorial Mechanics II	2nd semester	no semester recommended	-	0
Exercise: Mechanics II	2nd semester	no semester recommended	-	4

Module titel	Informatics in Civil Engineering (Compulsory subject)
Identifier	3011361
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2012
Valid until	-
Module level	Bachelor
Content	Introduction to computer science in civil engineering: overview and history; basics of computer based presentation and processing information, networks and internet; software architectures; databases; applications and tools of civil engineering; algorithms; data structures; recursion; flowcharts; basics of programming; Programming: introduction to an object oriented programming language; development environments; data structures and types; arrays; operators; expressions and statements; control structures; classes and objects; functions / methods; encapsulation; inheritance; polymorphism; exception handling; data input and output; graphical user interfaces
Learning Objectives/ Learning Outcomes	Knowledge about applications and tools for computer science in civil engineering; basic knowledge about information technology; understanding of modern software architectures; acquisition of selected algorithms for construction; learning the methodology for solving algorithmic problems based on an object-oriented programming language; ability for developing software applications related to engineering problems.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsfolien (in digitaler Form); Ergänzende Literatur
Language	German
Examination Terms	Graded written exam. Admission requirements for participation in the written exam: regular attendance at the small group exercises.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Universitätsprofessor Dr.-Ing. Jörg Blankenbach
ECTS Credits	3
Contact time (WSH)	3
Examination duration (min)	90
Total hours (h)	90,0
Contact hours (h)	45,0
Self-study hours (h)	45,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Small group exercise Introduction to Informatics in Civil Engineering and Programming (301136102)	1st semester	no semester recommended	0	2
Written exam Introduction to Informatics in Civil Engineering and Programming (301136101)	1st semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Exercise Introduction to Informatics in Civil Engineering and Programming	1st semester	no semester recommended	-	1

Module titel	Structural Design and Building Construction (Compulsory subject)
Identifier	3011363
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2008
Valid until	-
Module level	Bachelor/Master
Content	Einführung der Teilsicherheitsbeiwerte, Einführung in den Lastabtrag und die Lastweiterleitung verschiedener Tragelemente, Detailausbildung verschiedener Dachtragwerke, Vorstellung konstruktiver Details in Zusammenhang mit der Ableitung und Zerlegung unterschiedlicher Tragsysteme, Grundlagen der Bemessung im Hochbau, Berechnung einfacher Mauerwerks- und Holzbauteile, Vorstellung von Detaillösungen an den Schnittstellen unterschiedlicher Tragglieder, Aussteifungskonzepte und Gesamtstabilität
Learning Objectives/ Learning Outcomes	Erkennen der Zusammenhänge der Tragwerkelemente im Bauwesen; Aufstellung der Lastannahmen und Ermittlung der maßgebenden Lastfälle; Grundlagenwissen zum semi-probabilistischen Sicherheitskonzept; Fähigkeit zur Aufstellung statischer Berechnungen und Ausbildung der zugehörigen Details; Bemessung von Bauteilen aus Mauerwerk nach dem vereinfachten Verfahren; Grundlagenwissen zur Ausbildung von Treppen; Grundlagenwissen im Lastabtrag verschiedener Deckenkonstruktionen; Grundlagen zur Stabilisierung von Hochbauten
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Arbeitsunterlagen vom Lehrstuhl
Language	German
Examination Terms	Graded written exam. Admission requirements for participation in the written exam: passed term paper.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Univ.-Prof. Dr.-Ing. Martin Claßen
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	90
Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

- Specific Basics of Engineering and Civil ...
- + Structural Design and Building Construction (3011363)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Structural Design and Building Construction (301136301)	3rd semester	no semester recommended	0	0
Written exam Structural Design and Building Construction (301136302)	3rd semester	no semester recommended	5	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Tutorial: Structural design and building construction	3rd semester	no semester recommended	-	2
Lecture: Structural design and building construction	3rd semester	no semester recommended	-	2

Module titel	Building Physics (Compulsory subject)
Identifier	3011368
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2013
Valid until	-
Module level	Bachelor
Content	Klima und Mensch: Grundlagen Klimakunde; Grundlagen Klimawirkung; Wärme: Grundlagen der Wärmeleitung, Wärme- und Stoffübertragung und Wärmestrahlung; Thermische Kenngrößen; Thermisches Verhalten von Räumen und Außenbauteilen; Wärmebrücken; Instationäre Wärmeleitung in Bauteilen, Mechanismus der Wärmespeicherung; Energieeinsparungspotentiale; Feuchte: Feuchtetechnische Grundbegriffe; Wasserdampfgehalt der Luft, Wasserdampfpartialdruck, Tautemperatur, Diffusionswiderstand, Flüssigkeitsleitung, hx-Diagramm; Feuchtetransport durch Diffusion, Kapillardruck und strömende Luft; Vermeidung von Oberflächentauwasser; Glaser-Verfahren und dessen Grenzen; Schall: Wahrnehmung und Messung von Schall, Rechnen mit Schallpegeln; Schallschutz; Raumakustik; Luft- und Trittschalldämmung; Akustische Phänomene; Licht: Lichttechnische Grundbegriffe; Tageslicht im Freien und in Räumen, Tageslichtquotient, Beleuchtungsstärkeverteilung in Räumen; Praktische Anforderungen; Sonne und Himmel, Sonnenstand, Besonnungsdauer; Brandschutz: Brandschutzziele; Klassifizierung von Baustoffen und Bauteilen
Learning Objectives/ Learning Outcomes	Nach der Teilnahme an der Modulveranstaltung sind die Studierenden in der Lage bauphysikalische Phänomene aus den Bereichen Wärme, Feuchte, Tageslicht, Brandschutz, Schall zu verstehen und zu berechnen. Des Weiteren werden bauphysikalische Anforderungen beherrscht und somit können einfache Problemstellungen erkannt und gelöst werden. Studierende erlangen Kenntnisse über relevante normative Vorschriften.
(Study-Specific) Prerequisites	None
(recommended) Requirements	Knowledge of physics is strongly recommended, therefore participation in the module 'Physics' is recommended.
References	Vorlesungsunterlagen sowie die dort genannte Literatur, beispielsweise: Richter, Fischer, Jenisch, Freymuth, Stohrer, Häupl, Homann: Lehrbuch der Bauphysik - Schall - Wärme - Feuchte - Licht - Brand - Klima; Vieweg+Teubner, Wiesbaden (2008).
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the exam.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Universitätsprofessor Dr.-Ing. habil. Christoph van Treeck
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	90

- Specific Basics of Engineering and Civil ...
- + Building Physics (3011368)

Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam: Building Physics (301136801)	2nd semester	no semester recommended	5	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Übung: Bauphysik	2nd semester	no semester recommended	-	2
Vorlesung: Bauphysik	2nd semester	no semester recommended	-	2

Module titel	Structural Engineering I (Compulsory subject)
Identifier	3015569
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2017
Valid until	-
Module level	Bachelor
Content	Modeling and structural design for frame structures; System identification and idealization of the structure under consideration of support condition and hinges, as soon as actions from loads and displacements; Calculation of stress and deformation resultants for statically determinate and indeterminate systems; Structural analysis with the force method; Principal of virtual work for the calculation of stress and deformation resultants; Calculation of discrete displacements and rotations for the bending line; Influence line for force and displacement boundary conditions; Practical use of framework programs.
Learning Objectives/ Learning Outcomes	At the end of the course, the students are able to evaluate the load-bearing behavior of statically determined and indeterminate frame structures. They can calculate internal forces, discrete deformations and influence lines of complex static systems. The students are able to use framework programs for these tasks and they gained the ability to solve tasks in the area of structural analysis independently and to check the solutions plausibility.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	K. Meskouris, E. Hake: Statik der Stabtragwerke, Springer Verlag; K. Meskouris, C. Butenweg, E. Hake, S. Holler: Baustatik in Beispielen; Übungsumdruck; Vorlesungsumdruck
Language	German
Examination Terms	Graded written exam. There are no prerequisites for participation in the written exam. With the help of coluntary test certificates, points can be acquired once, which can be credited to the examination performance to a maximum extent of 20%. The exact criteria for the acquisition of bonus points are specified in the CMS.
Miscellaneous	-
Module coordinator	Universitätsprofessor Dr.-Ing. habil. Sven Klinkel
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	80
Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

- Specific Basics of Engineering and Civil ...
- + Structural Engineering I (3015569)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Structural Engineering I (301556901)	3rd semester	no semester recommended	5	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Exercise Structural Engineering I	3rd semester	no semester recommended	-	4

Module titel	Building Materials Science (Compulsory subject)
Identifier	3012176
Version	Angelegt über RWTH API als 1
Duration (Semester)	three semesters
Cycle (Semester)	winter semester
Valid from	Winter semester 2012
Valid until	-
Module level	Bachelor
Content	Baustoffkunde 1: Physikalische und chemische Grundlagen der Werkstoffkunde (Bindungsarten, Bindungsenergie, Plastizität, Phasendiagramme, Wärmedehnung und -leitfähigkeit, Dichte, Verformungseigenschaften, Spannungs-Dehnungsdiagramme, Grundlagen der Verbundwerkstofftheorie, Bruchmechanik); Metallische Werkstoffe: Stahl/ Aluminium; Werkstoffeigenschaften, Bewehrungsstahl, Prüfung, Korrosion; Baustoffkunde 2: Beton: Ausgangsstoffe und Werkstoffeigenschaften, Spannungs-Dehnungslinien in Abhängigkeit der Festigkeit, Werkstoffkorrosion, Werkstoffprüfung, Sonderbetone (Faserbeton, SVB, Hochleistungsbeton, Leichtbeton, Sichtbeton); Baustoffkunde 3: Mauerwerk: Wandkonstruktionen, Tragfähigkeits- und Verformungsverhalten, bauphysikalische Eigenschaften, Dauerhaftigkeit, Mauersteinarten und Verbundverhalten, Rissicherheit von Putzen; Kunststoffe: Verformungsverhalten, Gebrauchseigenschaften, Prüfung, Prinzipien der Herstellung, Struktur, Instandsetzungsmaterialien, Dauerhaftigkeit; Holz: Struktur, Trag- und Verformungsverhalten, physikalische Eigenschaften, Holzwerkstoffe, Holzschädigung durch Pilze und Insekten, Holzschutz; Glas: Anwendungsbeispiele, Trag- und Verformungsverhalten, physikalische Eigenschaften
Learning Objectives/ Learning Outcomes	Baustoffkunde 1: Grundsätzliches Verständnis für den Zusammenhang zwischen Bindungseigenschaften und Festigkeit; Verständnis für die Abläufe bei der Werkstoffverformung; Materialverhalten von Beton und Metallen als Grundlage für die Bemessung von Stahlbetonbauteilen; Grundsätze der Randbedingungen der Metallkorrosion für die konstruktive Durchbildung; Baustoffkunde 2: Kenntnisse über die Herstellung von Bauteilen aus Beton; Kenntnisse über das Verformungs- und Bruchverhalten von Beton als Grundlage für die Bemessung von Stahlbetonbauteilen; Kenntnisse über Verwendungsmöglichkeiten und Anwendungsgrenzen von Beton; Baustoffkunde 3: Kenntnisse über Arten, Formen und Herstellung von Mauerwerk-, Kunststoff-, Glas- und Holzbauteilen; Kenntnisse über die Einflüsse auf die Baustoffwiderstände (Tragfähigkeit und Verformung) von Mauerwerk, Kunststoff, Glas und Holz als Voraussetzung für die Bemessung; Kenntnisse über Verwendungsmöglichkeiten und Anwendungsgrenzen von Mauerwerk, Kunststoffen, Glas und Holz/Holzwerkstoffen
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdrucke zu Baustoffkunde 2 und Baustoffkunde 3; Zilch, K., Diederichs, C. J., Katzenbach, R.: Handbuch für Bauingenieure. 2002. Springer-Verlag Berlin Heidelberg New York. ISBN 3-540-65760-6
Language	German

– Specific Basics of Engineering and Civil ...
+ Building Materials Science (3012176)

Examination Terms	Graded written exams. The modul grade is weighted according to the CP distribution. There are no admission requirements for attending the written exams.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Univ.-Prof. Dr.-Ing. Thomas Matschei
ECTS Credits	9
Contact time (WSH)	8
Examination duration (min)	-
Total hours (h)	270,0
Contact hours (h)	120,0
Self-study hours (h)	150,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam Building Materials Science 1 (301217601)	1st semester	no semester recommended	4	0
Written exam Building Materials Science 2 (301217602)	2nd semester	no semester recommended	3	0
Written exam Building Materials Science 3 (301217603)	3rd semester	no semester recommended	2	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Tutorial Building Materials Science 3	3rd semester	no semester recommended	-	2
Lecture Building Materials Science 2	2nd semester	no semester recommended	-	2
Lecture Building Materials Science 1	1st semester	no semester recommended	-	2
Tutorial Building Materials Science 2	2nd semester	no semester recommended	-	1
Tutorial Building Materials Science 1	1st semester	no semester recommended	-	1

Module titel	Dynamics (Compulsory subject)
Identifier	3010809
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2012
Valid until	-
Module level	Bachelor
Content	Kinematics of point masses and rigid bodies; Dynamics (kinetics) of point masses, basics of kinetics of rigid bodies; Introduction of single- and multi-degree-of-freedom systems.
Learning Objectives/ Learning Outcomes	Familiarity with basic relations of Newton's Dynamics; ; Learning of basic dynamic methods and basic understanding of applying those methods to practical problems in civil engineering
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	D. Gross, W. Hauger, J. Schröder, W. Wall: Technische Mechanik 3
Language	German
Examination Terms	Graded written exam. There are no prerequisites for participation in the written exam. With the help of voluntary test certificates, points can be acquired once, which can be credited to the examination performance to a maximum extent of 20%. The exact criteria for the acquisition of bonus points are specified in the CMS
Miscellaneous	-
Module coordinator	Universitätsprofessor Dr.-Ing. habil. Sven Klinkel Priv.-Doz. Dr.-Ing. habil. Okayay Altay
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	80
Total hours (h)	120,0
Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam Dynamics (301080901)	4th semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture Dynamics	4th semester	no semester recommended	-	2
Tutorial Dynamics	4th semester	no semester recommended	-	2

— Specific Basics of Engineering and Civil ...
+ Introduction to Civil Engineering / Civil Engineering and ...

Module titel	Introduction to Civil Engineering / Civil Engineering and Society (Compulsory subject)
Identifier	3010734
Version	Angelegt über RWTH API als 1
Duration (Semester)	two semesters
Cycle (Semester)	winter/summer semester
Valid from	Summer semester 2015
Valid until	-
Module level	Bachelor
Content	Introduction to Civil Engineering: The course introduces to the various disciplines of Civil Engineering on the basis of a real construction project. The students will deal with problems and solutions of a modern construction project. Based on interactive blended learning concepts, the course will focus on intelligibly documenting a construction project. It will also show which engineering skills are particularly required to solve the tasks and problems of such a project and which legal, social and societal issues associated with it. Thereby, they receive a realistic picture of the requirements of their future work. Civil Engineering and Society: The students will be introduced to the fundamental aspects and issues in the context of "Engineering and Society". Motivated by the Sustainable Development Goals (SDGs), central topics and theories of sustainability, engineering ethics and social responsibility are introduced. Subsequently, the relevance of the interface between engineering sciences and society is clarified by addressing selected social structures in the field of gender equality policy, diversity as well as an international development policy. Examples for the implementation of a socially responsible and sustainable engineering design are discussed on the basis of mobility, urban planning as well as water supply topics. The teaching and learning concept of the lecture is based on a flipped classroom principle. The students work through the learning content independently and reflect on what they have learned with the help of various (online) reflection offerings. Discussion and reflection sessions at the beginning of the course, between the individual learning blocks and at the end form the framework of the course and serve the exchange in the plenum. Through the involvement of student initiatives, students gain insights into the practical implementation of the topics covered in theory.
Learning Objectives/ Learning Outcomes	Introduction to Civil Engineering: Gaining substantial understanding of engineering tasks to create and consolidate an understanding of the requirements of the methodical procedure of projects in Civil Engineering. Civil Engineering and Society: After completion of the course: The students • know the Sustainable Development Goals (SDGs) in the context of sustainability • understand the connection between sustainability and responsibility as well as the concomitant relevance of technology and innovation • understand the intersection of technology science and society as well as the resulting implications • understand the relevance of societal responsibility and identify fields of action in their respective occupational area • understand the social construction of technology and technical processes • understand the relevance of social and civic engagement • understand the relevance of technology and innovation within cultural and social structures

– Specific Basics of Engineering and Civil ...

+ Introduction to Civil Engineering / Civil Engineering and ...

	- discuss current issues with their fellow students and develop a broad understanding for cooperative learning - reflect the learned content and by writing a substantively rigorous and argumentatively logical assessment of a given issue.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded written exam 'Civil Engineering and Society' and ungraded cumulative e-tests 'Introduction to Civil Engineering'. The module grade is weighted according to the CP distribution. There are no prerequisites for taking the examination.
Miscellaneous	-
Module coordinator	Prof. Dr. phil. Carmen Leicht-Scholten
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	0
Total hours (h)	120,0
Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam: Civil Engineering and Society (301073401)	2nd semester	no semester recommended	2	0
e-Tests: Fundamentals in Civil Engineering (301073402)	1st semester	no semester recommended	2	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture: Civil Engineering and Society	2nd semester	no semester recommended	-	2
Lecture: Fundamentals in Civil Engineering	1st semester	no semester recommended	-	2

Module titel	Geotechnical Engineering I (Compulsory subject)
Identifier	3010811
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2008
Valid until	-
Module level	Bachelor/Master
Content	Determination of soil properties in the field and in the laboratory and classification of soils; water in the soil; stresses and deformations in the soil; settlement calculation; shear strength of soils; earth pressure and earth resistance determination; excavation construction.
Learning Objectives/ Learning Outcomes	Ability to derive qualitative soil properties from a given soil structure; ability to qualitatively describe the expected stress-strain behaviour of soil under a given load and description of the structure-soil interaction; mastery of the fundamentals of soil mechanics for determining the ultimate limit states and the serviceability limit states for selected applications in foundation engineering
(Study-Specific) Prerequisites	None
(recommended) Requirements	Recommended Knowledge: Mechanics I.
References	Vorlesungsumdruck "Geotechnik I"; Weiterführende Literatur: Gudehus: "Bodenmechanik"; Kolymbas: "Geotechnik - Bodenmechanik, Grundbau und Tunnelbau"; Kuntsche: "Geotechnik"; Schmidt: "Grundlagen der Geotechnik"; Simmer: "Grundbau 1 - Bodenmechanik und erdstatische Berechnungen"; Grundbautaschenbuch (Teil 1 und 2); Ziegler: "Geotechnische Nachweise nach EC 7 und DIN 1054: Einführung mit Beispielen"; Zeitschrift "Geotechnik"
Language	German
Examination Terms	Benotete Klausurarbeit. Voraussetzung für die Teilnahme an der Klausurarbeit: bestandene Hausarbeit.
Miscellaneous	-
Module coordinator	Universitätsprofessor Dr. Raul Fuentes
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	75
Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Geotechnical Engineering I (301081102)	3rd semester	no semester recommended	0	0
Written exam (or oral exam) Geotechnical Engineering I (301081101)	3rd semester	no semester recommended	5	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and exercise Geotechnical Engineering I	3rd semester	no semester recommended	-	4

Module titel	Water Quality Management and Principles of Sanitary Engineering (Compulsory subject)
Identifier	3015662
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2018
Valid until	-
Module level	Bachelor
Content	This course presents essential natural science and engineering fundamentals. It shows relationships, dependencies and how changes in one field of urban water management affect other fields of urban water management and/or water quality management. In water quality management the basics of limnology, different kinds of water utilisation, effects of wastewater discharges to a water body, the water quality classification and the ecological status according to EU Water Framework Directive are taught. Topics such as hygiene in urban areas (water supply, urban drainage), drainage comfort (flood protection, closing water loops on site), water protection (wastewater treatment, ways of discharge into water bodies) are discussed concerning urban water management. In addition to that the structure of water management is described and an overview of the water law is given.
Learning Objectives/ Learning Outcomes	On successfully completing this course unit, students will be able... ...to explain the most important scientific principles and processes as well as the interrelationships of water quality management and urban water management. ...to assess the self-purification processes in a water body. ...to determine basic data for the dimensioning of urban water management systems. ...to evaluate analysis results of relevant parameters for water quality and urban water management.
(Study-Specific) Prerequisites	None
(recommended) Requirements	None.
References	Course notes and additional literature are available on RWTHmoodle. Guger (2007): Siedlungswasserwirtschaft, 3. Auflage, Springer Verlag, Berlin, Heidelberg, ISBN: 978-3-540-34330-1
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the written exam.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Thomas Wintgens
ECTS Credits	3
Contact time (WSH)	2
Examination duration (min)	60
Total hours (h)	90,0

- Specific Basics of Engineering and Civil ...
 + Water Quality Management and Principles of Sanitary Engineering ...

Contact hours (h)	30,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Water Quality Management and Principles of Sanitary Engineering (301566201)	4th semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Vorlesung und Übung Grundlagen der Gewässergüte- und Siedlungswasserwirtschaft	4th semester	no semester recommended	-	2

Module titel	Structural Concrete I (Compulsory subject)
Identifier	3011369
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2009
Valid until	-
Module level	Bachelor
Content	Basic concepts of the design of concrete structures; load-bearing behavior of reinforced concrete as a composite material; semi-probabilistic safety concept; Ultimate Limit State (ULS) design for bending with axial force, shear and torsion forces; dimensioning of girders; simplified method for bending moment redistribution; Serviceability Limit States (SLS); basic principles of reinforcement layout and structural detailing
Learning Objectives/ Learning Outcomes	The course aims to convey basic knowledge of the bracing, conception and load-bearing behavior of reinforced concrete structures. By the end of the course, students should be able to safely design and dimension reinforced concrete sections for bending, axial force, shear force and torsion and be able to perform basic structural detailing.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdruck Massivbau I
Language	German
Examination Terms	Graded written exam. Admission requirements for participation in the written exam: passed term paper.
Miscellaneous	-
Module coordinator	Prof. Dr.-Ing. Martin Claßen
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	120
Total hours (h)	120,0
Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Structural Concrete I (301136901)	5th semester	no semester recommended	0	0
Written exam Structural Concrete I (301136902)	5th semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and exercise Structural Concrete I	5th semester	no semester recommended	-	4

Module titel	Methodology of Planning (Compulsory subject)
Identifier	3011364
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2008
Valid until	-
Module level	Bachelor
Content	<u>Institute of Transport Science</u> • Basics in track guided systems • Queuing processes in transportation • Analysis of traffic samples • Basics in economic efficiency calculation • Basics in timetable system <u>Institute of Urban and Transport Planning</u> • Essential human needs and functions and resulting infrastructure requirements • demographic and societal development as an input factor for planning • fundamentals of travel demand modelling • planning hierarchy and structure in Germany • error of human cognition und limits of planning <u>Institute of Highway Engineering</u> • Basic knowledge about the highway infrastructure Germany • Basic knowledge about driving dynamic and road design ; • Basic knowledge about noise and pollutant dispersion
Learning Objectives/ Learning Outcomes	• Basic understanding of the structure of the planning system (space and transport) in Germany • Basic knowledge of the work and planning process • Basic methodological knowledge in the fields of spatial planning and transport infrastructure ; • Basic knowledge of the rail mode of transport • Basic knowledge about the planing processes in the area of highways
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	-
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the exam.
Miscellaneous	-
Module coordinator	Universitätsprofessor Dr.-Ing. Tobias Kuhnimhof, Universitätsprofessor Dr.-Ing. Nils Nießen, Universitätsprofessor Dr.-Ing. habil. Markus Oeser ;
ECTS Credits	4

— Specific Basics of Engineering and Civil ...
+ Methodology of Planning (3011364)

Contact time (WSH)	4
Examination duration (min)	120
Total hours (h)	120,0
Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam Methodology of Planning (301136401)	3rd semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and exercise Methodology of Planning	3rd semester	no semester recommended	-	4

Module titel	Project and Contract Management (Compulsory subject)
Identifier	3026711
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2022
Valid until	-
Module level	Bachelor
Content	The module provides comprehensive knowledge of project management in construction. The perspectives of the of the various project participants in the planning and construction process and the basics of the legal framework. legal framework conditions are taught. - Project management - Cost-schedule quality from the client's point of view - Basics of tendering and awarding - Basics of building law and contracts - Cost-schedule quality from the contractor's point of view - Basics of supplementary management - Insight into the digitalisation of the construction industry and Building Information Modelling.
Learning Objectives/ Learning Outcomes	The work of engineers is a project-oriented discipline. The aim of the course is to learn all the necessary skills of the construction project business in order to be able to move confidently in this working environment. After successful completion of the module, the students have comprehensive knowledge of construction projects so that they can carry out the planning and control of costs, deadlines and quality from the perspective of the client or an engineering service provider as well as from the perspective of a contractor on the construction side. They are able to think through a construction project from the start to the end of the project, taking into account the respective boundary conditions, and to independently set up, implement and control projects of low complexity. The students are familiar with the common tendering and award procedures for engineering and construction services and know which requirements under public procurement law must be observed. must be observed. They are familiar with the most important types of construction contracts. They understand the interdependencies that exist between the topics of construction contracts, project objectives and tendering. Based on this knowledge, the students are able to process cases of supplementary management. The students have understood the importance of technical building regulations and the contents of building permit procedures.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	-

– Specific Basics of Engineering and Civil ...
+ Project and Contract Management (3026711)

Language	German
Examination Terms	Graded written exam. Prerequisite for participation in the written exam: passed term paper.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Katharina Klemmt-Albert
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Project and Contract Management (302671101)	3rd semester	no semester recommended	5	0
Term Paper Project and Contract Management (302671102)	3rd semester	no semester recommended	0	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and exercise Project and Contract Management	3rd semester	no semester recommended	-	4

Module titel	Construction Site Management 1 (Compulsory subject)
Identifier	3010819
Version	V2
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2022
Valid until	-
Module level	Bachelor/Master
Content	Erdbau/Tiefbau (Fertigungsverfahren, Baumaschinen, Geräteauswahl, Leistungsabstimmung, Kalkulation); Baugruben (verfahrenstechnische Aspekte); Betonbau (Schalung, Rüstung, Bewehrung, Betonherstellung und -verarbeitung); Hebezeuge.
Learning Objectives/ Learning Outcomes	Die Studierenden erwerben Kenntnisse über die Verfahrenstechniken im Erd- und Spezialtiefbau. Sie erwerben die Fähigkeit, Leistungsgeräte zu kalkulieren. Die Studierenden erlangen Kenntnisse über die Verfahrenstechniken im Betonbau. Die Studierenden erwerben Grundkenntnisse über die baubetriebliche Abwicklung von Betonbaustellen.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Lecture reprint Construction Site Management 1
Language	German
Examination Terms	Graded written exam. Admission requirement for participation in the written exam: passed term paper.
Miscellaneous	-
Module coordinator	Dr.-Ing. Robert Hartung
ECTS Credits	3
Contact time (WSH)	2
Examination duration (min)	-
Total hours (h)	90,0
Contact hours (h)	30,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Construction Site Management 1 (301081902)	5th semester	no semester recommended	0	0
Exam Construction Site Management 1 (301081901)	5th semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Tutorial Construction Site Management 1	5th semester	no semester recommended	-	2

Module titel	Urban and Regional Planning I (Compulsory subject)
Identifier	3013846
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2009
Valid until	-
Module level	Bachelor
Content	History of urban development; urban development and influential factors; legal foundations, procedures and planning processes in regional and state planning as well as in regional and urban planning and urban land use planning; dimensioning and calculation bases for urban planning
Learning Objectives/ Learning Outcomes	Students should acquire the ability to understand the interrelationships of spatial planning in general as well as the planning system of the Federal Republic of Germany and become familiar with the basic methods, procedures and instruments of spatial planning.
(Study-Specific) Prerequisites	None
(recommended) Requirements	Knowledge from the course "Planning Methodology" and "Spatial Development and Transport Policy" is recommended for participation in this module.
References	Vorlesungsumdruck Stadt- und Regionalplanung I; Übungsmaterialien
Language	German
Examination Terms	Graded written exam. The prerequisite for participation in the written exam: passed project work.
Miscellaneous	-
Module coordinator	Dr.-Ing. Andreas Witte
ECTS Credits	3
Contact time (WSH)	3
Examination duration (min)	60
Total hours (h)	90,0
Contact hours (h)	45,0
Self-study hours (h)	45,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Project Thesis Urban and Regional Planning I (301384601)	5th semester	no semester recommended	0	0
Exam Urban and Regional Planning I (301384602)	5th semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Vorlesung und Übung Stadt- und Regionalplanung I	5th semester	no semester recommended	-	3

Module titel	Surveying (Compulsory subject)
Identifier	3011362
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2013
Valid until	-
Module level	Bachelor
Content	Geodetic coordinate systems (geocentric GPS coordinates, Gauss-Krueger coordinates, UTM coordinates), height systems (NN-heights, NHN-heights, ellipsoid heights); measuring units; two- and three-dimensional surveying, modeling and mapping of natural and artificial objects (topography, tachymeter and satellite based (GPS) topographic surveying, determining longitudinal and cross profiles, calculation methods for coordinates, areas and volumes, height determination by leveling and trigonometric methods); optical and sensory basics of geodetic instruments (digital levels, electro-optical distance measurements and tachymeter, rotating lasers, GPS receiver, tilt sensors); mapping by photogrammetry and scanning methods; staking out and monitoring of buildings; positioning and navigation in road, rail, tunnel, bridge and waterway constructions (calculation and staking out of route elements like straights, circles and clothoids); optical and laser-based plumbing and alignment; deformation measurements and settlement determination
Learning Objectives/ Learning Outcomes	Recognize the scope and necessary quality of existing or novel planning documents; ability to decide on the required measurement methods including the achievable measurement accuracy and the appropriateness of the measurement method; being familiar with the coordinate and height calculation methods including the evaluation of correctness; reliable evaluation of the survey results and the planning documents
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Übungsumdrucke; Lehrbuch Witte/Schmidt: Vermessungskunde und Grundlagen der Statistik für das Bauwesen, Wichmann Verlag
Language	German
Examination Terms	Graded written exam. Admission requirements for participation in the written exam: regular attendance and regular attendance at the small group exercises, passed term paper.
Miscellaneous	-
Module coordinator	Universitätsprofessor Dr.-Ing. Jörg Blankenbach
ECTS Credits	3
Contact time (WSH)	3
Examination duration (min)	120
Total hours (h)	90,0

– Specific Basics of Engineering and Civil ...
+ Surveying (3011362)

Contact hours (h)	45,0
Self-study hours (h)	45,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exercise: Surveying (301136204)	2nd semester	no semester recommended	0	2
Small group exercise Surveying (301136203)	2nd semester	no semester recommended	0	0
Term paper Surveying (301136202)	2nd semester	no semester recommended	0	0
Written exam Surveying (301136201)	2nd semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture: Surveying	2nd semester	no semester recommended	-	1

Module titel	Internship Surveying (Compulsory subject)
Identifier	3013947
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2008
Valid until	-
Module level	Bachelor
Content	Height and positional determination of points, lines and areas by means of geodetic measuring methods (surveying leveling); staking out a predetermined road route to nature by using conventional and satellite-based measuring instruments; surveying of longitudinal and cross profiles for determination of soil volumes; topographic surveying using tachymeter and satellite-based measurement methods; preparing a site plan in a given scale, including representation of the terrain by contour lines.
Learning Objectives/ Learning Outcomes	Ability for surveying a parcel of land related to horizontal and vertical issues; capability to carry out simple leveling and position measurements for the purposes of building site projects; knowledge about the professional documentation of measurement results; ability for assessing the achievable accuracy and the required effort of surveying services related to time and costs.
(Study-Specific) Prerequisites	None
(recommended) Requirements	Knowledge from the course 'Surveying' is strongly recommended.
References	Übungsumdrucke; Lehrbuch Witte/Schmidt: Vermessungskunde und Grundlagen der Statistik für das Bauwesen, Wichmann Verlag
Language	German
Examination Terms	The examination consists of a graded term paper / result documentation. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	Universitätsprofessor Dr.-Ing. Jörg Blankenbach
ECTS Credits	2
Contact time (WSH)	2
Examination duration (min)	0
Total hours (h)	60,0
Contact hours (h)	30,0
Self-study hours (h)	30,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Internship Surveying (301394702)	2nd semester	no semester recommended	0	2
Term paper Surveying (301394701)	2nd semester	no semester recommended	2	0

- Specific Basics of Engineering and Civil ...
- Construction Project Life Cycle: Design ...
- + Introduction to Finite Element Method for Structural Analysis ...

Module titel	Introduction to Finite Element Method for Structural Analysis (Compulsory elective subject)
Identifier	3019369
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2020
Valid until	Summer semester 2024
Module level	Bachelor
Content	- Energy methods and the principle of virtual displacements - Isoparametric concept - Variational formulation of finite elements - Finite elements for trusses and beams - Locking and alternative FE formulations - Linear equation solvers
Learning Objectives/ Learning Outcomes	At the end of the course, you can - understand the structure of a FE program. - create your own FE program. - code your own functions in Matlab. - use energy methods and the principle of virtual displacements. - transform differential equations into the weak form. - develop your own finite elements. - understand the cause of locking-effects. - modify element formulations to reduce locking effects. - understand the way solvers for linear equation systems work and know their limitations.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdrucke
Language	German
Examination Terms	The examination consists of a graded written exam and a graded term paper. The grade is made up of 80% of the written exam and 20% of the paper. Prerequisite for participation in the written exam: passed term paper.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. habil. Sven Klinkel
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	120,0

- Specific Basics of Engineering and Civil ...
- Construction Project Life Cycle: Design ...
- + Introduction to Finite Element Method for Structural Analysis ...

Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term Paper Introduction to Finite Element Method for Structural Analysis (301936901)	5th semester	no semester recommended	0	0
Exam Introduction to Finite Element Method for Structural Analysis (301936902)	5th semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Exercise Introduction to Finite Element Method for Structural Analysis	5th semester	no semester recommended	-	4

- Specific Basics of Engineering and Civil ...
- Integral Planning or Introduction to ...
- + Integral Planning (3028324)

Module titel	Integral Planning (Compulsory elective subject)
Identifier	3028324
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2023
Valid until	-
Module level	Bachelor/Master
Content	The course is designed to provide students with basic knowledge and methodological skills for the planning of building structures. The focus is on the early service phases in the project. The focus is on the necessary interaction and communication with the project participants and the impact of planning decisions on the aspects of functionality, costs and sustainability. The contents of the module are divided into 2 core areas: General methodological competencies and exemplary implementation through business game / presentation. Within the lecture, students are taught the following theoretical basics: • General: From the idea to the description of requirements - Sequence and early phases of a building construction project and its interfaces. • Introduction to the preparation of utilization concept and requirements specification • Introduction to various digital and parameterizable planning methods and their possible applications in the project • Implications and subject of integral planning / early exchange of all involved sub-trades as an essential part of integral planning • Leadership of interdisciplinary knowledge transfer, reduction of conflict potential • Communication and cooperation - various aspects of professional exchange in project work • Complexity reduction in communication and project organization in cooperation The acquired basics are applied in a business game or group work. In doing so, the students deal with the following thematic focal points in an exemplary manner: • Collaborative solution of exemplary tasks/problems of planning in early performance phases. • Prerequisites, components, interfaces (information technology) • Creation of structural concept and material selection • Location and dimension of technical centers on the basis of an energy concept • Exemplary route planning / coordination with other trades • Iterative coordination with other technical planners • Dealing with planning changes • Contract design with BIM, effects on conventional contract models and project examples
Learning Objectives/ Learning Outcomes	After completing the module, students should be able to: • understand what integral planning means in the construction industry and what role it plays, especially in early service phases • select working methods appropriate to the project requirements • understand the interdependencies between the different trades • recognize and implement the need for communication and coordination between trades • be able to communicate complex, trade-specific content to other project participants • Identify project requirements from the client's point of view and derive planning requirements from them, taking into account sustainability and cost aspects. • know methods/tools for integral planning in early phases of work and to be able to apply them exemplarily.

- Specific Basics of Engineering and Civil ...
- Integral Planning or Introduction to ...
- + Integral Planning (3028324)

	- to develop an understanding of methods, constraints/conflicts and interfaces between trades - put oneself in other roles and to be able to work in a team as well as independently/ integral planning
(Study-Specific) Prerequisites	-
(recommended) Requirements	Recommended knowledge: Project &; Contract Management, Structural Engineering, Introduction to Civil Engineering
References	Documents from lectures and exercises
Language	German
Examination Terms	The examination consists of a graded presentation ("Referat"). The grade is composed of 50% written work and 50% oral presentation. There are no prerequisites for taking the exam.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Katharina Klemt-Albert
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	120,0
Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Seminar Business Simulation Integral Planning (302832401)	5th semester	5th semester	0	0
Exam Integral Planning (302832402)	5th semester	5th semester	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Exercise Integral Planning	5th semester	5th semester	-	4

- Specific Basics of Engineering and Civil ...
- Integral Planning or Introduction to ...
- + Introduction to Finite Element Method for Structural Analysis ...

Module titel	Introduction to Finite Element Method for Structural Analysis (Compulsory elective subject)
Identifier	3019369
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2023
Valid until	-
Module level	Bachelor
Content	- Energy methods and the principle of virtual displacements - Isoparametric concept - Variational formulation of finite elements - Finite elements for trusses and beams - Locking and alternative FE formulations - Linear equation solvers
Learning Objectives/ Learning Outcomes	At the end of the course, you can - understand the structure of a FE program. - create your own FE program. - code your own functions in Matlab. - use energy methods and the principle of virtual displacements. - transform differential equations into the weak form. - develop your own finite elements. - understand the cause of locking-effects. - modify element formulations to reduce locking effects. - understand the way solvers for linear equation systems work and know their limitations.
(Study-Specific) Prerequisites	-
(recommended) Requirements	-
References	Vorlesungsumdrucke
Language	German
Examination Terms	The examination consists of a graded written exam and a graded term paper. The grade is made up of 80% of the written exam and 20% of the paper. Prerequisite for participation in the written exam: passed term paper.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. habil. Sven Klinkel
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	120,0

- Specific Basics of Engineering and Civil ...
- Integral Planning or Introduction to ...
- + Introduction to Finite Element Method for Structural Analysis ...

Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term Paper Introduction to Finite Element Method for Structural Analysis (301936901)	5th semester	no semester recommended	0	0
Exam Introduction to Finite Element Method for Structural Analysis (301936902)	5th semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Exercise Introduction to Finite Element Method for Structural Analysis	5th semester	no semester recommended	-	4

Module titel	Digital Building and Construction (Compulsory elective subject)
Identifier	3026088
Version	-
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2022
Valid until	-
Module level	Bachelor
Content	Digitalization is making its way into the construction industry. Through a well-grounded education in the field of Digital Building and Construction, graduates will occupy a new area of activity, which is the focus of the whole construction industry. The module conveys knowledge of digital methods in construction industry and deals intensively with the Building Information Modeling method. Specifically the following course content will be covered: Building Information Modeling (BIM) - BIM as a method of project management - digital tools in the construction industry and their application Information Technology -Prerequisites, components, interfaces -Link to geodesy Methodology and processes -Process and document management -workflow management Communication and collaboration -reduction of complexity and project organization Contract design -contract design with BIM, impact on conventional contract models Project examples
Learning Objectives/ Learning Outcomes	After successful completion of the module, students will have deepened knowledge of the use of digital methods in construction projects. They can use the Building Information Modeling method. The students have a comprehensive picture of the Construction Industry 4.0 and are able to make cross-relationships between the conventional construction industry and the conventional project management.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	General knowledge in construction project management, fundamental knowledge in CAD-/BIM-usage

– Institute Internship
+ Digital Building and Construction (3026088)

References	Albert. A (Hrsg). (2020). Schneider Bautabellen für Ingenieure. mit Berechnungshinweisen und Beispielen. BIM (Building Information Modeling). Bundesanzeiger Verlag. ISBN 978-3-8462-1140-3; Borrmann, A., König, M., Koch, C., Beetz, J. (2021). Building Information Modeling. Technologische Grundlagen und industrielle Praxis. Springer Vieweg. ISBN 978-3-658-33360-7; Richtlinien aus der Reihe VDI 2552 Building Information Modeling (BIM); Weitere Literaturhinweise gibt es in der Vorlesung.
Language	German
Examination Terms	The exam consists of a graded term paper (group work). There are no prerequisites for taking the exam.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Katharina Klemmt-Albert
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Digital Building and Construction (302608801)	6th semester	no semester recommended	5	4

Module titel	Institute Internship: Railway Engineering (Compulsory elective subject)
Identifier	3022320
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Einführung in die Stellwerkstechnik an der Eisenbahntechnischen Lehr- und Versuchsanlage (ELVA); Vertiefung der Kenntnisse durch praktische Übungen an der ELVA; Problemstellung: Fahrstraßenbildezeiten und -auflösezeiten; Einführung in LUKS; Fahrplanverifizierung an der ELVA; Abweichung vom Regelbetrieb (Praktische Übungen an der ELVA); Störungen (Praktische Übung an der ELVA); Betriebsdisposition.
Learning Objectives/ Learning Outcomes	Grundkenntnisse im Eisenbahnsicherungswesen; Fähigkeiten zur Durchführung von Fahrlagenplanung, Trassenmanagement und Betriebsführung; Erwerben praktischer Kenntnisse in den genannten Bereichen.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/result documentation. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Nils Nießen
ECTS Credits	5
Contact time (WSH)	3
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	45,0
Self-study hours (h)	105,0

— Institute Internship

+ Institute Internship: Railway Engineering (3022320)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Railway Engineering (302232001)	6th semester	no semester recommended	5	3

Module titel	Institute Internship: Hydromechanics (Compulsory elective subject)
Identifier	3022325
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Analyse wissenschaftlicher Fachveröffentlichungen; wasserbauliche Natur- und Labormessungen; Darstellung der Messergebnisse in schriftlicher und mündlicher Form; Einzel- und Gruppenarbeit.
Learning Objectives/ Learning Outcomes	In der Institutspraktikumsphase erhalten die Studierenden einen Einblick in das eigenständige wissenschaftliche Arbeiten und in forschungsorientierte Arbeitsweisen. Die Studierenden vertiefen ihr Verständnis hydromechanischer Prozesse durch Anschauung und eigene praktische Erfahrungen. Weiterhin sollen die Studierenden Erfahrungen mit der Analyse von wissenschaftlichen Fachveröffentlichungen (journal papers) sammeln. Durch den Umgang mit Messtechnik entwickeln die Studierenden die Fähigkeit zur Konzeption und Durchführung von Experimenten und/oder Naturmessungen. Darüber hinaus lernen die Studierenden die Ergebnisse wissenschaftlicher Untersuchungen auf wissenschaftliche Weise sowohl schriftlich als auch mündlich darzustellen. Damit sollen Kompetenzen und Rüstzeuge zur zeiteffektiven und hochqualitativen Bearbeitung von wissenschaftlichen Untersuchungen (z.B. Bachelorarbeiten) geschaffen werden.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper (or a graded report on the internship) and a graded presentation. 50% of the module grade is derived from the grade of the term paper (or internship report) and 50% from the grade of the presentation. Prerequisites for participation in the examination: compulsory attendance of the internship and compulsory participation in the safety briefing.
Miscellaneous	-
Module coordinator	-
ECTS Credits	5
Contact time (WSH)	5
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	75,0

— Institute Internship

+ Institute Internship: Hydromechanics (3022325)

Self-study hours (h)

75,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Hydromechanics (302232501)	6th semester	no semester recommended	5	5

Module title	Institute Internship: Structural Analysis (Compulsory elective subject)
Identifier	3030184
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2024
Valid until	-
Module level	Bachelor
Content	Use of modern 3D scanning and modelling techniques on naturally grown unprocessed wood; modelling with framework programs; generation and calculation of simple structural analyses.
Learning Objectives/ Learning Outcomes	Basic knowledge of structural analysis. In-depth knowledge of performing, automating and evaluating parametric 3D modelling in addition to self-generated 3D scans. In-depth knowledge of modelling and calculation with bar structure programs, as well as training and calculation of the design of connections. Evaluation of the results by comparison with self-performed measurements.
(Study-Specific) Prerequisites	-
(recommended) Requirements	Structural Engineering I, Steel Structures I, Introduction to Finite Element Method for Structural Analysis
References	-
Language	German
Examination Terms	The exam consists of a graded term paper/documentation of results and a graded presentation/colloquium. The module grade is 50% based on the grade of the term paper/documentation of results and 50% based on the grade of the presentation/colloquium. Prerequisite for participation in the examination is compulsory attendance at the internship.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Prof. Dr.-Ing. habil. Sven Klinkel
ECTS Credits	5
Contact time (WSH)	2
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	30,0
Self-study hours (h)	120,0

— Institute Internship

+ Institute Internship: Structural Analysis (3030184)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Structural Analysis (303018401)	6th semester	5th semester	5	2

Module titel	Institute Internship: Structural Analysis and Dynamics (Compulsory elective subject)
Identifier	3022322
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Geschwindigkeits- und Beschleunigungsmessungen an realen Bauwerken; Auswertung von Messdaten; Modellbildung mit Stabwerksprogrammen; Modellkalibrierung; Programmierung von Methoden zur Signalverarbeitung.
Learning Objectives/ Learning Outcomes	Grundlegende Kenntnisse der digitalen Signalverarbeitung; Eingehende Kenntnisse in der Durchführung, Auswertung und Analyse von Schwingungsmessungen an Bauwerken; Modellierungsstrategien für die numerische Untersuchung baudynamischer Fragestellungen und Beurteilung der Ergebnisse durch Abgleich mit selbst durchgeführten Messungen.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/documentation and a graded presentation/colloquium. 50% of the module grade is based on the grade of the term paper/documentation and 50% on the grade of the presentation/colloquium. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	-
ECTS Credits	5
Contact time (WSH)	2
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	30,0
Self-study hours (h)	120,0

— Institute Internship

+ Institute Internship: Structural Analysis and Dynamics (3022322)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Structural Analysis and Dynamics (302232201)	6th semester	no semester recommended	5	2

Module titel	Institute Internship: Building Materials (Compulsory elective subject)
Identifier	3022323
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Vorstellen eines Vorhabens mit komplexen Materialprüfungen und der durchzuführenden Versuche; Mitarbeit bei der Gewinnung und Vorbereitung von Probekörpern; Mitarbeit bei der Versuchsdurchführung; Mitarbeit bei der Bewertung.
Learning Objectives/ Learning Outcomes	Die Veranstaltung soll den Studierenden einen Einblick in die praktische Umsetzung wissenschaftlicher Arbeit vermitteln. Dabei sollen grundlegende Kenntnisse in der Materialprüfung erlangt werden, um diese bei Fragestellungen der Praxis, wie z. B. der Bauwerksdiagnose oder der Umsetzung baustofftechnologischer Konzepte, anwenden zu können. Mit Abschluss der Veranstaltung sind die Studierenden in der Lage, Bewertungen von baustofftechnologischen Fragestellungen vorzunehmen und zu formulieren.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/documentation and a graded presentation/colloquium. 50% of the module grade is based on the grade of the term paper/documentation and 50% on the grade of the presentation/colloquium. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	-
ECTS Credits	5
Contact time (WSH)	6
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	90,0
Self-study hours (h)	60,0

— Institute Internship

+ Institute Internship: Building Materials (3022323)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Building Materials (302232301)	6th semester	no semester recommended	5	6

Module titel	Institute Internship: Engineer Meets User (Compulsory elective subject)
Identifier	3022324
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Recherche bestehender Studien und stadtplanerischer Grundlagen; Dokumentation von Beobachtungen und Interviews mit NutzerInnen vor Ort; Aufbereitung und Analyse der gewonnenen Daten; Vorbereitung und Durchführung eines Diskussionsworkshops mit PraktikerInnen und NutzerInnen; Transfer der Ergebnisse in Handlungsempfehlungen mit anschließender Präsentation.
Learning Objectives/ Learning Outcomes	Transfer zwischen ingenieurwissenschaftlichem Wissen und Diversität im Rahmen von Entwicklungen herstellen zu können; Auseinandersetzung mit Planungsrealität vor dem Hintergrund kommunalpolitischer Rahmenbedingungen; Grundkenntnisse in Ergebnissen und Diskursen sozialwissenschaftlich basierter Forschung zu Bedürfnissen, Anforderungen und Herausforderungen diverser NutzerInnengruppen zu erlangen und daraus Empfehlungen ableiten zu können; Verständnis, dass der Mensch und seine diversen Bedürfnisse im Zentrum technischer und wissenschaftlicher Innovation steht.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper and a presentation/colloquium. The module grade is determined by 75% of the grade of the term paper and 25% of the grade of the presentation/colloquium. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr. phil. Carmen Leicht-Scholten
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

— Institute Internship

+ Institute Internship: Engineer Meets User (3022324)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Engineer Meets User (302232401)	6th semester	no semester recommended	5	4

Module titel	Institute Internship: Geotechnics and Monitoring in Civil Engineering (Compulsory elective subject)
Identifier	3026086
Version	-
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2022
Valid until	-
Module level	Bachelor
Content	In der Institutspraktikumsphase besteht fakultativ die Möglichkeit, dass die Studierenden aktiv aktuelle Forschungsaufgaben des ausrichtenden Instituts kennenlernen und mitarbeiten. Sie erhalten dadurch einen Einblick in das eigenständige wissenschaftliche Arbeiten und erlernen forschungsorientierte Arbeitsweisen. Die Institute werden jeweils für eine eingeschränkte Anzahl von Praktikumsplätzen ein derartiges Praktikum anbieten und betreuen. Institutspraktikum „Geo- und Messtechnik im Bauwesen“: Vorstellung laufender Projekte und Forschungsvorhaben sowie der damit zusammenhängenden Versuchsdurchführungen und Auswertung. Eigenständige Durchführung fachspezifischer bzw. forschungsorientierter Versuche und Aufgaben sowie deren Auswertung wie z.B.: Bestimmung und Messung von Behaglichkeitskriterien; Wärmestrommessung und Speicherefähigkeit aktivierter Bauteile; Elektrotechnik; Direkte und indirekte Aufschlüsse zur Baugrunderkundung; Bestimmung der Eigenschaften von Boden und Fels; Untersuchung des Verbundverhalten von Böden und Geokunststoffen; Versuche zu geothermischen Fragestellungen; Versuche zu verfahrenstechnischen Grundlagen und Optimierungen von geotechnischen Anwendungen. Fakultativ besteht einer eingeschränkten Anzahl an Studierenden die Möglichkeit einer tiefgehenden Auswertung und Dokumentation der Forschungsaufgaben.
Learning Objectives/ Learning Outcomes	Grundkenntnisse über Fragestellungen zum Bauwesen in der Forschung und Praxis; Grundlagenkenntnisse zur Programmierung von eigenen Messwerkzeugen mit einem Arduino; Grundlagenkenntnisse zur Baugrunderkundung und Geotechnik; Grundlagenkenntnisse über die Durchführung von Versuchen; Fähigkeit zur Durchführung von Versuchsauswertungen; Einblick ins eigenständige wissenschaftliche Arbeiten und forschungsorientierte Arbeitsweisen erlernen.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	Es werden Kenntnisse aus „Energie und Gebäudetechnik“, „Geotechnik I“, Interesse an Laborarbeit, Programmierung und Messtechnik empfohlen.
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/result documentation. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr. Christoph van Treeck; Univ.-Prof. Dr. Raul Fuentes
ECTS Credits	5
Contact time (WSH)	6

— Institute Internship

+ Institute Internship: Geotechnics and Monitoring in Civil ...

Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	90,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Geotechnics and Monitoring in Civil Engineering (302608601)	6th semester	no semester recommended	5	6

Module titel	Institute Internship: Engineering Hydrology (Compulsory elective subject)
Identifier	3022326
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Ausarbeitungen zu laufenden Forschungs- und Entwicklungsaufgaben aus dem Bereich der Ingenieurhydrologie Erlernen von grundlegenden Präsentationstechniken; Aufbau und Strukturierung von medienunterstützten Präsentationen und Selbstlernmedien;
Learning Objectives/ Learning Outcomes	Die Studierenden sollen anhand von konkreten Fragestellungen aus der Ingenieurhydrologie das eigenständige, selbstorganisierte Arbeiten erlernen. Dazu werden sie in laufende Forschungs- und Entwicklungsaufgaben eingebunden. Zum Abschluss der Praktikumsphase sollen die Studierenden die Fähigkeit erlangt haben, sich strukturiert und mit konkreten Zeitvorgaben in ein abgegrenztes Aufgabenfeld einzuarbeiten und aussagekräftige Präsentationen zu ihren Ausarbeitungen zu erstellen.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper (or a graded documentation of results) and a graded presentation. 70% of the module grade is based on the grade of the term paper (or the documentation of results) and 30% on the grade of the presentation. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Heribert Nacken
ECTS Credits	5
Contact time (WSH)	1
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	15,0
Self-study hours (h)	135,0

— Institute Internship

+ Institute Internship: Engineering Hydrology (3022326)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Engineering Hydrology (302232601)	6th semester	no semester recommended	5	1

Module titel	Institute Internship: Lightweight Construction (Compulsory elective subject)
Identifier	3022327
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Vorstellung von Leichtbaukonstruktionen und Darstellung der Besonderheiten; Mitarbeit bei der Konstruktion und Montage von Leichtbauelementen; Mitarbeit bei Vorbereitung und Durchführung von Versuchen; Mitarbeit bei der Auswertung von Messdaten.
Learning Objectives/ Learning Outcomes	In der Veranstaltung sollen die Studierenden theoretische Kenntnisse und praktische Erfahrungen in den Themenbereichen Konstruktion, Montage sowie messtechnische Untersuchungen von Leichtbaukonstruktionen erwerben.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/documentation and a graded presentation/colloquium. 50% of the module grade is based on the grade of the term paper/documentation and 50% on the grade of the presentation/colloquium. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	-
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

— Institute Internship

+ Institute Internship: Lightweight Construction (3022327)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Lightweight Construction (302232701)	6th semester	no semester recommended	5	4

Module titel	Institute Internship: Structural Concrete (Compulsory elective subject)
Identifier	3022328
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Vorstellen eines Forschungsvorhabens und der durchzuführenden Versuche; Mitarbeit bei der Herstellung der Versuchskörper; Mitarbeit bei der Versuchsdurchführung; Mitarbeit bei der Versuchsauswertung.
Learning Objectives/ Learning Outcomes	Das Institutspraktikum vermittelt den Studierenden einen Einblick in die wissenschaftliche Forschungstätigkeit. Es werden grundlegende Kenntnisse über die Herstellung von Bauteilen aus Stahl- und Spannbeton oder aus nicht-metallischer Bewehrung erlangt und praktische Erfahrungen bei der Durchführung und -auswertung von wissenschaftlichen Versuchen gesammelt.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/documentation and a graded presentation/ colloquium. 50% of the module grade is based on the grade of the term paper/documentation and 50% on the grade of the presentation/colloquium. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Martin Claßen
ECTS Credits	5
Contact time (WSH)	6
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	90,0
Self-study hours (h)	60,0

— Institute Internship

+ Institute Internship: Structural Concrete (3022328)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Structural Concrete (302232801)	6th semester	no semester recommended	5	6

Module titel	Institute Internship: Sanitary Engineering and Management of Municipal Solid Waste (Compulsory elective subject)
Identifier	3022329
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	In this practical course on urban water and waste management (institute practical course), students implement fundamentals of urban water management and/or apply them in the lab. The course consists of three parts. In part 1, students dimension a stormwater basin or tank and conduct a sensitivity analysis for the dimensioning procedure. In part 2, they assess water quality at a wastewater treatment plant. In part 3, they run a series of practical laboratory tests, including both laboratory analysis (e.g., sum parameters, biological parameters, water hardness) and tests on bench-scale plants for water and wastewater treatment technologies (e.g., activated carbon, sand filtration, membrane filtration).
Learning Objectives/ Learning Outcomes	On successfully completing this course unit, students will be able... ...to dimension stormwater basin or tank according to the currently applicable regulations. ...to conduct a sensitivity analysis and identify critical parameters. ...to evaluate water quality data of a wastewater treatment plant, and assess the data according to their applicable limit values and significance. In doing so, they can identify, classify and evaluate the influence of e. g. dry and rainy weather conditions. ...to name the relevant (waste)water quality parameters and describe the associated laboratory analysis procedures. ...to name relevant water treatment processes and describe their mode of operation as well as their most important parameters. ...to carry out environmental-analytical as well as process-technical laboratory tests, evaluate them and classify the results. ...to present acquired knowledge and findings in scientific form and communicate them orally and in writing.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	Arbeitsblatt ATV-DVWK-A 198 (2003): Vereinheitlichung und Herleitung von Bemessungswerten für Abwasseranlagen, Hennef Arbeitsblatt DWA-A 102 (BWK-A/M 3) (2020): Grundsätze zur Bewirtschaftung und Behandlung von Regenwetterabflüssen zur Einleitung in Oberflächengewässer Arbeitsblatt DWA-A 131 (2016): Bemessung von einstufigen Belebungsanlagen, Hennef Crittenden et al. (2012) MWH's Water Treatment: Principles and Design, 3rd edition, John Wiley & Sons, ISBN: 9781118131473 Deutsche Einheitsverfahren zur Wasser-, Abwasser- und Schlammuntersuchung - Physikalische, chemische, biologische und bakteriologische Verfahren, Loseblattsammlung, Herausgeber Wasserchemische Gesellschaft in der GDCh, DIN Gujer (2007) Siedlungswasserwirtschaft, 3. Auflage, Springer Verlag, Berlin, Heidelberg, ISBN: 978-3-540-34330-1 Metcalf & Eddy (2013) Wastewater Engineering: Treatment and Resource Recovery, 5th edition, McGraw-Hill Education, ISBN: 978-0073401188

— Institute Internship

+ Institute Internship: Sanitary Engineering and Management of ...

Language	German
Examination Terms	The examination consists of a graded internship report (including evaluations). Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Thomas Wintgens
ECTS Credits	5
Contact time (WSH)	5
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	75,0
Self-study hours (h)	75,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Sanitary Engineering and Management of Municipal Solid Waste (302232901)	6th semester	no semester recommended	5	5

Module titel	Institute Internship: Urban Planning (Compulsory elective subject)
Identifier	3022331
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Durchführung von Erhebungen im Verkehrswesen (Zählungen, Messungen, Videobeobachtungen, Befragungen etc.) sowie Aufbereitung und Darstellung von Daten; Bestandsaufnahme, -analyse stadtplanerischer Strukturen (Situationsanalyse, Stärken- und Schwächen-Analyse); Begleitung von Projekten, Teilnahme an Sitzungen und Ausschüssen; Laborversuche zur Straßenbautechnik; EDV-gestützte Planungsmethoden in der Stadt- und Verkehrsplanung.
Learning Objectives/ Learning Outcomes	Einführung in die Arbeitsfelder und Arbeitsweisen von Verkehrsingenieuren; Methoden der Datenerhebung und -aufbereitung in Verkehrswesen und Stadtplanung (Zählungen, Messungen, Videobeobachtungen, Befragungen etc.); Anwendung von Software im Verkehrswesen.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/result documentation. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	-
ECTS Credits	5
Contact time (WSH)	3
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	45,0
Self-study hours (h)	105,0

— Institute Internship

+ Institute Internship: Urban Planning (3022331)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Urban Planning (302233101)	6th semester	no semester recommended	5	3

Module titel	Institute Internship: Road Engineering (Compulsory elective subject)
Identifier	3022332
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Durchführung von Erhebungen im Verkehrswesen (Zählungen, Messungen, Videobeobachtungen, Befragungen etc.) sowie Aufbereitung und Darstellung von Daten; Begleitung von Projekten, Teilnahme an Sitzungen und Ausschüssen; Laborversuche zur Straßenbautechnik; Teilnahme an Fahrsimulationen; Laborversuche zum Tunnelbetrieb; EDV-gestützte Planungsmethoden in der Stadt- und Verkehrsplanung.
Learning Objectives/ Learning Outcomes	Einführung in die Arbeitsfelder und Arbeitsweisen von Verkehrsingenieuren; Methoden der Datenerhebung und -aufbereitung in Verkehrswesen und Straßenbautechnik; Anwendung von Software im Verkehrswesen; Anwendung von Labor- und In-situ-Prüfverfahren in der Straßenbautechnik.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/result documentation. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	Prof. Dr. Alvaro Garcia-Hernandez
ECTS Credits	5
Contact time (WSH)	3
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	45,0
Self-study hours (h)	105,0

— Institute Internship

+ Institute Internship: Road Engineering (3022332)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Road Engineering (302233201)	6th semester	no semester recommended	5	3

Module title	Institute Internship: Design of Concrete Structures (Compulsory elective subject)
Identifier	3022333
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Anwendung von Software zur Bemessung von Bauteilen; Anwendung eines CAD Programms zur Erstellung von Bewehrungszeichnungen.
Learning Objectives/ Learning Outcomes	Im Praktikum sammeln die Studierenden praktische Erfahrungen in der Anwendung von FEM-Software und Zeichenprogrammen. Anhand praxisnaher Beispiele lernen die Studierenden exemplarisch die rechnergestützte Berechnung und Bemessung ausgewählter Bauteile und die Umsetzung der Ergebnisse in Bewehrungspläne. Im Rahmen der Hausübung werden die erlernten Fähigkeiten und Methoden anhand eines Beispielgebäudes vertieft und zusammengeführt.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/documentation of results and a graded colloquium. The module grade is determined by 80% of the grade of the term paper/documentation and 20% of the grade of the colloquium. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Martin Claßen
ECTS Credits	5
Contact time (WSH)	3
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	45,0
Self-study hours (h)	105,0

— Institute Internship

+ Institute Internship: Design of Concrete Structures (3022333)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Design of Concrete Structures (302233301)	6th semester	no semester recommended	5	3

Module titel	Institute Internship: Design of Steel Structures (Compulsory elective subject)
Identifier	3022334
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Anwendung von Software zur Bemessung von ; Stahlbaukonstruktionen; ;Anwendung eines CAD Programms zur Erstellung von Ausführungsplänen im Stahlbau.
Learning Objectives/ Learning Outcomes	In der Veranstaltung Praktikum soll den Studierenden die sichere Anwendung praxisrelevanter Software zur Bemessung und Konstruktion von Stahlbaukonstruktionen vermittelt werden. Verschiedene Bemessungs- und Zeichenprogramme werden hierbei praktisch angewendet. Anhand konkreter Beispiele werden die typischen Arbeitsschritte einer Tragwerksplanung im Stahlbau durchgeführt, ausgehend von einer Bemessung der einzelnen Bauteile hin zur Erstellung von Ausführungszeichnungen.
(Study-Specific) Prerequisites	Prerequisites for participation in the examination: passed modules Mathematics I, Mathematics II, Mechanics I, Mechanics II.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded term paper/documentation and a graded presentation/ colloquium. 50% of the module grade is based on the grade of the term paper/documentation and 50% on the grade of the presentation/colloquium. Prerequisite for participation in the examination: compulsory attendance of the practical training.
Miscellaneous	-
Module coordinator	-
ECTS Credits	5
Contact time (WSH)	3
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	45,0
Self-study hours (h)	105,0

— Institute Internship

+ Institute Internship: Design of Steel Structures (3022334)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Institute Internship: Design of Steel Structures (302233401)	6th semester	no semester recommended	5	3

Module titel	Structural Engineering II (Compulsory elective subject)
Identifier	3010810
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2009
Valid until	-
Module level	Bachelor
Content	Internal force calculation for statically determinate systems in 3D Force method in 3D Calculation of deformations in 3D Methods to determine geometric indeterminacy Structural analysis using the direct stiffness method Calculation of influence lines with the direct stiffness method Fundamentals of structural dynamics, dynamic stiffness and mass matrix Calculation of natural frequencies and mode shapes.
Learning Objectives/ Learning Outcomes	At the end of this course, students are able to determine the internal forces and displacements of statically determinate and indeterminate structures in 3D and can assess the load-bearing capacity and deformation of geometrically indeterminate structures. They can determine natural frequencies and mode shapes of frame structures and can formulate the (reduced) differential equation of motion. Students have acquired the fundamentals needed for further structural engineering courses.
(Study-Specific) Prerequisites	None
(recommended) Requirements	Knowledge from „Structural Engineering I“ is recommended. This includes: Simplifications and the recognition of statically determinate subsystems Pole diagrams and the principle of virtual work Determination of equivalent spring stiffnesses Influence lines Symmetry and antisymmetry
References	K. Meskouris, E. Hake: Statik der Stabtragwerke, Springer Verlag; K. Meskouris, C. Butenweg, E. Hake, S. Holler: Baustatik in Beispielen; Übungsumdruck; Vorlesungsumdruck
Language	German
Examination Terms	Graded written exam. There are no prerequisites for participation in the written exam. Bonus points can be acquired once accompanying the lecture. They can be credited to the examination performance to a maximum extent of 20%. The exact criteria for the acquisition of bonus points are specified in the CMS.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Universitätsprofessor Dr.-Ing. habil. Sven Klinkel Priv.-Doz. Dr.-Ing. habil. Okay Altay
ECTS Credits	3
Contact time (WSH)	3
Examination duration (min)	80
Total hours (h)	90,0

Contact hours (h)	45,0
Self-study hours (h)	45,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Structural Engineering II (301081002)	4th semester	no semester recommended	0	0
Written exam (or oral exam) Structural Engineering II (301081001)	4th semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Vorlesung und Übung Baustatik II	4th semester	no semester recommended	-	3

Module titel	Introduction to Materials Mechanics (Compulsory elective subject)
Identifier	3010820
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2011
Valid until	-
Module level	Bachelor
Content	Vorstellung verschiedener Materialverhaltensweisen (z.B. Stahl, Beton, Asphalt); Diskussion von experimentellen Ergebnissen; Erarbeitung von ein- und mehrdimensionalen Spannungs-Dehnungs-Zusammenhängen; Einbettung in die Grundgleichungen der Mechanik (Kinematik, Gleichgewicht, Materialgesetz); Numerische Berechnung einfacher Strukturen; Vergleich Experiment - Simulation, Parameteridentifikation; Praktikum zu Selberrechnen; Umgang mit kommerziellen Programmsystemen
Learning Objectives/ Learning Outcomes	Grundlegendes Verständnis verschiedener Materialverhaltensweisen (elastoplastisch, viskoelastisch usw.); Kenntnis der Grundgleichungen der Mechanik (Kinematik, Gleichgewicht, Materialgesetz) in drei Dimensionen; Sicherheit in der Anwendung leistungsstarker numerischer Methoden; Verständnis grundlegender Vorgehensweisen in der Werkstoffmechanik: experimentelle Beobachtung, Modellierung, Simulation, Parameteridentifikation
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdruck; Handzettel der Übungen
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the exam.
Miscellaneous	-
Module coordinator	Modulverantwortliche: Universitätsprofessorin Dr.-Ing. Stefanie Reese
ECTS Credits	3
Contact time (WSH)	3
Examination duration (min)	90
Total hours (h)	90,0
Contact hours (h)	45,0
Self-study hours (h)	45,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam (or oral exam) Introduction to Materials Mechanics (301082001)	5th semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Tutorial: Introduction to Materials Mechanics	5th semester	no semester recommended	-	3

Module titel	Structural Concrete II (Compulsory elective subject)
Identifier	3011373
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter/summer semester
Valid from	Summer semester 2014
Valid until	-
Module level	Bachelor
Content	Design and construction of concrete slabs, columns and walls; bracing of structural systems; design and construction of consoles and footings; introduction to prestressed concrete and pre-dimensioning of prestressed concrete elements
Learning Objectives/ Learning Outcomes	The course aims to familiarize students with the design of structures and their components. By the end of the course, students should be able to safely design and dimension reinforced concrete structures and have obtained an introduction to prestressed concrete.
(Study-Specific) Prerequisites	None
(recommended) Requirements	Knowledge of 'Methodology of Planning' is recommended for participation.
References	Vorlesungsumdruck Massivbau I; Vorlesungsumdruck Massivbau II; Übungsumdruck Massivbau II
Language	German
Examination Terms	Graded written exam. Admission requirements for participation in the written exam: ungraded passed term paper and colloquium.
Miscellaneous	-
Module coordinator	Prof. Dr.-Ing. Martin Claßen
ECTS Credits	4
Contact time (WSH)	3
Examination duration (min)	120
Total hours (h)	120,0
Contact hours (h)	45,0
Self-study hours (h)	75,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Structural Concrete II (301137301)	6th semester	no semester recommended	0	0
Written exam Structural Concrete II (301137302)	6th semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and exercise Structural Concrete II	6th semester	no semester recommended	-	3

Module titel	Steel Structures I/II (Compulsory elective subject)
Identifier	3010822
Version	Angelegt über RWTH API als 1
Duration (Semester)	two semesters
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2009
Valid until	-
Module level	Bachelor
Content	Stahlbau I: Eigenschaften des Baustoffes Stahl; Grundlagen des Sicherheitskonzeptes des Eurocode 3; Querschnittsklassifizierung; Bemessung von einfachen Stahlbaukonstruktionen; Entwurf und Bemessung von Anschlüssen; Konstruktive Gestaltung von Anschlussdetails; Stahlbau II: Ermüdungsgerechtes Konstruieren; Stabilität; Berechnung von Tragwerken nach Theorie II. Ordnung
Learning Objectives/ Learning Outcomes	Stahlbau I: Verständnis für das Tragverhalten des Baustoffes Stahl; Verständnis des Sicherheitskonzeptes für Stahlkonstruktionen; Grundkenntnisse zur Zerlegung von Tragwerken in für die Nachweise relevante Einzelbauteile und Einzelnachweise; Sicheres Bemessen von Stahlquerschnitten für die Beanspruchung aus Biegung, Längskraft und Querkraft; Sicheres Bemessen von Anschlussdetails (Schweiß- und Schraubverbindungen); Grundkenntnisse der konstruktiven Gestaltung von geschweißten und geschraubten Anschlussdetails; Anfertigen von einfachen Ausführungszeichnungen / -skizzen; Stahlbau II: Bemessung und Entwurf von komplexeren Stahlbaukonstruktionen; Maßgebende Stabilitätsfälle und Grundkenntnisse der zugehörigen Bemessungsregeln; Lösung von Stabilitätsproblemen nach Theorie II. Ordnung; Kenntnisse zur Stahlermüdung
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdrucke Stahlbau I + II; DIN-Normen; EN-Normen; Fachbücher
Language	German
Examination Terms	Graded written exams. The module grade is weighted according to the CP distribution. Prerequisite for the participation in the written exams are passed corresponding homeworks. As an alternative to the homework, term papers can be done according to the chair's requirements after consultation with the chair.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Universitätsprofessor Dr.-Ing. Markus Feldmann
ECTS Credits	8
Contact time (WSH)	6
Examination duration (min)	-
Total hours (h)	240,0

Contact hours (h)	90,0
Self-study hours (h)	150,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Homework assignment Steel Structures II (301082202)	6th semester	no semester recommended	0	0
Homework assignment Steel Structures I (301082201)	5th semester	no semester recommended	0	0
Written exam (or oral exam) Steel Structures I (301082203)	5th semester	no semester recommended	4	0
Written exam (or oral exam) Steel Structures II (301082204)	6th semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Vorlesung und Übung Stahlbau II	6th semester	no semester recommended	-	3
Vorlesung und Übung Stahlbau I	5th semester	no semester recommended	-	3

Module titel	Wastewater Management (Compulsory elective subject)
Identifier	3020956
Version	V1
Duration (Semester)	two semesters
Cycle (Semester)	winter/summer semester
Valid from	Summer semester 2020
Valid until	-
Module level	Bachelor
Content	Wastewater Treatment: This course deals with the basics of wastewater treatment. The course covers the principles of waste water treatment and the functionalities of the treatment stages of a conventional waste water treatment plant as well as their dimensioning. Urban Drainage: This course deals with urban drainage. The course focusses on the discharge of combined wastewater and stormwater as well as on the retention and treatment of flows before discharge into a water body.
Learning Objectives/ Learning Outcomes	Wastewater Treatment: On successfully completing this course unit, students will be able... ...to explain the basic principles of municipal wastewater treatment ...to understand the role of each unit process within typical treatment process trains and their interactions. ...to calculate the dimensions of a mechanical treatment train ...to calculate the dimensions of a biological treatment train in accordance to the DWA-A131 ...to present various processes for eliminating of nutrients and how they function ...to name alternative processes for wastewater treatment. Urban Drainage: On successfully completing this course unit, students will be able... ...to describe the runoff generation and to calculate the effective precipitation. ...to describe system components and the structure of different drainage systems. ...to explain the constructions and operating principles of stormwater tanks, retention soil filters and infiltration systems. ... to dimension stormwater tanks, retention soil filters and infiltration systems according to the relevant German regulations.
(Study-Specific) Prerequisites	None
(recommended) Requirements	Knowledge from 'Basics of Water Quality and Urban Water Management' is recommended for participation in this module.
References	Wastewater Treatment Arbeitsblatt DWA-A 131 (2016): Bemessung von einstufigen Belebungsanlagen, Hennef ATV-Handbuch Biologische und weitergehende Abwasserreinigung (1997), 4. Auflage, Ernst & Sohn, Berlin Gujer (2007): Siedlungswasserwirtschaft, 3. Auflage, Springer Verlag, Berlin, Heidelberg, ISBN: 978-3-540-34330-1 Imhoff et al. (2018): Taschenbuch der Stadtentwässerung, 32. Auflage, Vulkan-Verlag GmbH, Essen Urban Drainage:

– Water Engineering and Management
+ Wastewater Management (3020956)

	ATV-Handbuch Abwasserableitung (2006) Hrsg.: Deutsche Vereinigung für Wasserwirtschaft, Abwasser und Abfall e.V. Butler & Davies (2011): Urban Drainage, Spon Verlag, 3. Auflage, London Gujer (2007): Siedlungswasserwirtschaft, 3. Auflage, Springer Verlag, Berlin, Heidelberg, ISBN: 978-3-540-34330-1 Hager (1995): Abwasserhydraulik, Springer-Verlag, Berlin Imhoff et al. (2018): Taschenbuch der Stadtentwässerung, 32. Auflage, Vulkan-Verlag GmbH, Essen
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the written exam.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Univ.-Prof. Dr.-Ing. Thomas Wintgens
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	120,0
Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Wastewater Management (302095601)	6th semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and exercise Wastewater Treatment	6th semester	no semester recommended	-	2
Lecture and exercise Urban Drainage	5th semester	no semester recommended	-	2

Module titel	Sustainability Assessment (Compulsory elective subject)
Identifier	3020963
Version	V2
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2023
Valid until	-
Module level	Bachelor
Content	Nachhaltigkeitsbewertung Grundlagen: Überblick über europäisches und nationales Umweltrecht (Bund, Länder); Nachhaltigkeitsleitbilder/-indikatoren; Umweltqualitätsziele sowie Nachhaltigkeitsziele und -strategien; Umwelt-Auditing (EMAS, DIN EN ISO 14001 ff.); Grundlagen zum Aufbau und zur Implementierung von Umweltmanagementsystemen; Überblick über gängige Umwelt- und Nachhaltigkeitslabels; Einführung in das Ressourcenmanagement. Nachhaltigkeitsbewertung Methoden: Überblick über gängige Methoden der Nachhaltigkeitsbewertung; Methoden innerhalb des Umwelt-/ Nachhaltigkeitsmanagements; Einführung in die Ökobilanz und Risikobewertung; Stoffstrom- und Flächenmanagement; Einführung in die Nachhaltigkeitsbewertung (z.B. soziale Nachhaltigkeitsbewertung); Einführung in die Nachhaltigkeitsbewertung auf Unternehmensebene.
Learning Objectives/ Learning Outcomes	Nachhaltigkeitsbewertung Grundlagen: Ziel der Veranstaltung ist es, die elementaren Grundlagen des öffentlichen und betrieblichen Umwelt-/Nachhaltigkeitsmanagements, die normativen Anforderungen sowie Kenntnisse über Aufbau, Inhalt und Ziele der wichtigsten Managementsysteme im Nachhaltigkeitsbereich zu vermitteln und sie an ausgewählten Beispielen zu erproben. Die Studierenden erwerben vertiefte Kenntnisse der wissenschaftlichen Grundlagen des öffentlichen und betrieblichen Nachhaltigkeitsmanagements und der zugehörigen Prinzipien. Weiter erhalten sie Einblicke in das Rohstoffmanagement. Die Studierenden sind durch die Veranstaltung in der Lage die Umwelt-/Nachhaltigkeitsrelevanz öffentlicher und betrieblicher Entscheidungen sachkundig zu beurteilen. Nachhaltigkeitsbewertung Methoden: Ziel des Moduls ist es, die elementaren Grundlagen der Umwelt- und Nachhaltigkeitsbewertung zu vermitteln sowie Kenntnisse über Aufbau, Inhalt und Ziele der wichtigsten Methoden darzustellen und zu erörtern und sie an ausgewählten Beispielen zu erproben. Die Studierenden erwerben vertiefte Kenntnisse der wissenschaftlichen Grundlagen der methodischen Prinzipien und Instrumente sowie der zugehörigen Indikatoren. Die Studierenden sind durch die Veranstaltung in der Lage, die Umwelt-/Nachhaltigkeitsbewertung in vereinfachter Form anzuwenden und auf neue Produkte und Unternehmensbereiche zu übertragen, um hierüber eine Aussage über die Nachhaltigkeit treffen zu können.
(Study-Specific) Prerequisites	-
(recommended) Requirements	-
References	EMAS, DIN EN ISO 14001, Sustainable Development Goals of United Nations
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the written exam.
Miscellaneous	-

Module coordinator	Modulverantwortliche: Univ.-Prof. Dr.-Ing. Marzia Traverso
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	120,0
Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Sustainability Assessment (302096301)	5th semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture Sustainability Assessment	5th semester	no semester recommended	-	4

Module titel	Hydraulic Engineering (Compulsory elective subject)
Identifier	3020965
Version	V1
Duration (Semester)	two semesters
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2019
Valid until	-
Module level	Bachelor
Content	River Engineering: Hydrology and water resource management; water legislation and administration; river and stream control; flood control; water-retaining structures; barrages and weirs; design; structural elements; basis of computation; climate change; nature-orientated hydraulic engineering Dams and Hydropower: Dams: embankment dams; concrete dams; stability and serviceability; operation equipment; Hydropower: low, medium and high pressure plants; pumped storage hydro power stations
Learning Objectives/ Learning Outcomes	River Engineering: Students are introduced to the German water legislation as a framework for planning and permission of hydraulic engineering projects. The course is designed to review the main theoretical principles taught in Hydromechanics I in individual and group-related work. The unity of theory and practice is demonstrated. The students are encouraged to deal with technical complex structures. Apart from technical aspects, the course focuses on key issues of current political debates such as climate change and flood control. Dams and Hydropower: Students learn about the conceptual and structural design of dams, hydropower stations and other hydraulic-engineering structures. They become aware of the relevance of hydraulic structures in social context. Moreover the importance of normative frameworks in hydraulic engineering design is shown. Students develop the capability to organize, design and assess large hydraulic engineering structures. As a result, the participants of the course will be encouraged to deal with complex problems. The specific practical relevance and knowledge about the broad range of hydraulic engineering fields are essential elements of this course.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdruck; Naudascher, E. (1992): Hydraulik der Gerinne und Gerinnebauwerke. Springer. ISBN 3-211-82366-2; Strobl, T.; Zunic, F. (2006): Wasserbau – Aktuelle Grundlagen - Neue Entwicklungen. Springer. ISBN 978-3-540-22300-9; Vischer, D.; Huber, A. (1993): Wasserbau. Springer. ISBN 978-3-540-56178-1; Giesecke J.; Mosonyi, E. (2005): Wasserkraftanlagen – Planung, Bau und Betrieb. Springer. ISBN 3-540-44391-6; Kaczynski, J. (1994): Stauanlagen – Wasserkraftanlagen. Werner-Verlag. ISBN; Patt, H.; Jürging, P.; Kraus, W. (2004): Naturnaher Wasserbau – Entwicklung und Gestaltung von Fließgewässern. Springer. ISBN 978-3-540-20095-6
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the written exam.
Miscellaneous	-

Module coordinator	Modulverantwortlicher: Univ.-Prof. Dr.-Ing. Holger Schüttrumpf
ECTS Credits	6
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	180,0
Contact hours (h)	60,0
Self-study hours (h)	120,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Hydraulic Engineering (302096501)	5th semester	no semester recommended	6	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture River Engineering	5th semester	no semester recommended	-	2
Lecture Dams and Hydropower	4th semester	no semester recommended	-	2

Module titel	Water Resources Management and Hydrology (Compulsory elective subject)
Identifier	3020966
Version	V1
Duration (Semester)	two semesters
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2019
Valid until	-
Module level	Bachelor
Content	Wasserwirtschaft und Hydrologie I: Aufbau und Funktionsweise des Wasserhaushaltes; Grundlagen der Teilkompartimente Niederschlag, Verdunstung, Abfluss und Speicherung; Grundlagen der quantitativen und qualitativen Wasserwirtschaft; Grundlagen der Herleitung von Bemessungswerten in der Wasserwirtschaft (hydrologische Statistik); Anwendungsbeispiele aus der Wasserwirtschaft (Ausweisung von Retentionsflächen, Hochwasserschadenspotenzial-Analysen, Erosionsmodellierung, Speicherwirtschaft, DV-Aufgaben in der Hydrologie). Wasserwirtschaft und Hydrologie II: Grundlagen und Anwendung der Fließgewässermorphologie; Grundlagen und praxisrelevante Anwendung der Fließgewässertypologie; Wechselwirkungen von Abfluss und Gerinnemorphologie; Berechnungsgrundlagen des Strahlungshaushalts; Grundlagen der abiotischen und biotischen Gewässerkenntnis; Grundlagen und Anwendung des Energie- und Nährstoffhaushalts von Fließgewässern; Interaktion Gewässer - Grundwasser; Grundlagen des diffusen Stoffeintrages (vor dem Hintergrund der gesetzlichen Regelungen); Grundlagen der praxisrelevanten Anwendung der wasserwirtschaftlichen Maßnahmenplanung und Bewirtschaftungspläne.
Learning Objectives/ Learning Outcomes	Wasserwirtschaft und Hydrologie I: Die Studierenden sollen eine profunde Wissensbasis zu den Prozessabläufen des Wasserkreislaufes (Hydrologie) erhalten und die Zusammenhänge der qualitativen und quantitativen Wasserwirtschaft anhand von Anwendungsbeispielen erarbeiten. Dabei sollen die Studierenden lernen, eigenständig konkreten Aufgaben aus der Wasserwirtschaft zu lösen und ihr erarbeitetes Wissen im Rahmen des self-assessment fortlaufend überprüfen. Wasserwirtschaft und Hydrologie II: Die Studierenden sollen aufbauend auf der Wissensbasis aus der Lehrveranstaltung Wasserwirtschaft und Hydrologie I ein vertieftes Verständnis der wasserwirtschaftlichen Planung vor dem Hintergrund der nationalen und europaweiten gesetzlichen Regelungen erlangen und das Wissen selbstständig anhand von praxisrelevanten Anwendungsbeispielen umsetzen. Dabei sollen die Studierenden ihr erarbeitetes Wissen im Rahmen des self-assessment fortlaufend überprüfen.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	-
Language	German
Examination Terms	Graded written exams. The weighting is based on the distributions of CPs. There is no admission requirement for participation in the written exams.
Miscellaneous	-

- Water Engineering and Management
+ Water Resources Management and Hydrology (3020966)

Module coordinator	Modulverantwortlicher: Univ.-Prof. Dr.-Ing. Heribert Nacken
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	120,0
Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Water Resources Management and Hydrology I (302096601)	5th semester	no semester recommended	2	0
Exam Water Resources Management and Hydrology II (302096602)	6th semester	no semester recommended	2	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture Water Resources Management and Hydrology I	5th semester	no semester recommended	-	2
Lecture Water Resources Management and Hydrology II	6th semester	no semester recommended	-	2

Module titel	Introduction to Rock Mechanics and Tunnelling (Compulsory elective subject)
Identifier	3026087
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2022
Valid until	-
Module level	Bachelor/Master
Content	• Terminology • Basics of Rock Mechanics: structural model, anisotropy, water permeability, stress-strain behaviour, strength • Basics of determining rock mechanical parameters • Design principles for rock structures: cavities, slopes, etc. • Routing elements for tunnels • Tunnel statics: Kastner method, Schulze/Duddeck method, Kennlinien method • Open cut tunnels: Basic principles • Tunnels in closed construction: Excavation classes, shotcrete construction method (NÖT), etc. • Basics of blasting technology • Basics of mechanised tunnelling in soil and rock • Control measurements • Project examples
Learning Objectives/ Learning Outcomes	• Basic knowledge of the essential rock parameters and ability to describe them with simple rock mechanical models. • Knowledge of the most important construction methods and materials in rock construction • Ability to select a suitable tunnelling method depending on the existing underground • Knowledge and application of the basic tunnel static calculation models for the design of tunnel structures
(Study-Specific) Prerequisites	None
(recommended) Requirements	Knowledge of Mechanics I, Mechanics II, Hydromechanics I and passed term paper 'Geotechnical Engineering I' is strongly recommended for participation.
References	• Vorlesungs- und Übungsumdruck • einschlägige Normen und Empfehlungen • einschlägige Lehr- und Fachbücher: Müller-Salzburg, Wittke, Hudson/Harrison, Goodman, Maidl, Herrenknecht, Kolymbas, Muir Wood, Tunnelbau-Taschenbuch, etc. • einschlägige Fachzeitschriften: geotechnik, Int. Journal of Rock Mechanics & Mining Sciences, tunnel, geomechanics & tunnelling, Tunnelling and Underground Space Technology, etc. • Institutsveröffentlichungen • Tagungsberichte: DGGT, ISRM, STUVA, etc.
Language	German
Examination Terms	Graded written exam. Admission requirements for participation in the written exam: passed term paper.
Miscellaneous	-

- Construction Management and Geotechnical ...
 + Introduction to Rock Mechanics and Tunnelling (3026087)

Module coordinator	Akad. Dir. Dipl.-Ing. Martin Feinendegen
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term Paper Assignment Introduction to Rock Mechanics and Tunnelling (302608701)	4th semester	no semester recommended	0	0
Written Exam Introduction to Rock Mechanics and Tunnelling (302608702)	4th semester	no semester recommended	5	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture Introduction to Rock Mechanics and Tunnelling	4th semester	no semester recommended	-	2
Exercise Introduction to Rock Mechanics and Tunnelling	4th semester	no semester recommended	-	2

Module titel	Energy and Building Services Engineering (Compulsory elective subject)
Identifier	3020960
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2019
Valid until	-
Module level	Bachelor
Content	basics of energy supply; fossil and renewable energy sources, energy production and conversion, use in national and international contexts; political framework conditions, climate protection; support programs for the building sector; energy quality of buildings, determination of energy demand, energy savings and energy efficiency in the building sector, energy saving regulation; basics of sustainable building planning, efficient technologies (passive, active), renewable energies for buildings; building services engineering: fundamentals of heating and ventilation technology, heating load calculation, overview of different heating systems, thermal protection during summer, cooling load calculation, overview of electrical, conducting and sanitary technology
Learning Objectives/ Learning Outcomes	Students will gain background knowledge about the global framework conditions with regard to energy supply; get to know fossil and renewable energy sources in terms of extraction and energy conversion; discuss political frameworks and developments / goals with regard to climate protection, as well as gain elementary background knowledge for assessing the energetic quality of buildings. The lecture section building services engineering focuses on the necessary basics of heating and ventilation technology and the necessary calculation rules for heat load and cooling load calculation, and provides a concise overview of the electrical, control and sanitary technology.
(Study-Specific) Prerequisites	None
(recommended) Requirements	The successful completion of the module Building Physics is recommended.
References	Vorlesungsmaterialien
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the written exam.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Univ.-Prof. Dr.-Ing. Christoph van Treeck
ECTS Credits	5
Contact time (WSH)	3
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	45,0

- Construction Management and Geotechnical ...
- + Energy and Building Services Engineering (3020960)

Self-study hours (h) 105,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Energy and Building Services Engineering (302096001)	5th semester	no semester recommended	5	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture Energy and Building Services Engineering	5th semester	no semester recommended	-	3

Module titel	Geotechnical Engineering II (Compulsory elective subject)
Identifier	3012184
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter/summer semester
Valid from	Summer semester 2022
Valid until	-
Module level	Bachelor
Content	Plasticity solutions - upper / lower bound; Shallow foundations; Deep foundations; Slope failure; Complex combinations; Groundwater management.
Learning Objectives/ Learning Outcomes	Ability to identify and develop simple plasticity solutions applied to geotechnical engineering problems; Experience in applying the most important computational verifications for foundation engineering constructions and slope stability; Awareness of the differences of realistic, more complex, regarding stability analysis; Experience in designing simplified groundwater management systems for various applications.
(Study-Specific) Prerequisites	Admission requirement for participation in the written exam: passed term paper Geotechnical Engineering I.
(recommended) Requirements	Recommended knowledge: Mechanics I, Geotechnical Engineering I.
References	• Vorlesungsumdruck "Geotechnik II" Weiterführende Literatur: • Bachus: "Grundbaupraxis" • Kolymbas: "Geotechnik - Bodenmechanik, Grundbau und Tunnelbau" • Kuntsche: "Geotechnik" • Schmidt: "Grundlagen der Geotechnik" • Simmer: "Grundbau 2 - Baugruben und Gründungen" • Grundbautaschenbuch (Teil 3) • Ziegler: "Geotechnische Nachweise nach EC 7 und DIN 1054: Einführung mit Beispielen" • Zeitschrift "Geotechnik"
Language	German
Examination Terms	Graded written exam. Admission requirements for participation in the written exam: passed term paper "Geotechnical Engineering II". With the help of voluntary test certificates, points can be acquired once, which can be credited to the examination performance to a maximum extent of 20%. The exact criteria for the acquisition of bonus points are specified in the CMS.
Miscellaneous	-
Module coordinator	Universitätsprofessor Dr. Raul Fuentes
ECTS Credits	5
Contact time (WSH)	4
Examination duration (min)	75

Total hours (h)	150,0
Contact hours (h)	60,0
Self-study hours (h)	90,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
term paper Geotechnical Engineering II (301218401)	4th semester	no semester recommended	0	0
written exam (or oral exam) Geotechnical Engineering II (301218402)	4th semester	no semester recommended	5	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Vorlesung und Übung: Geotechnik II	4th semester	no semester recommended	-	4

Module titel	Construction Site Management 2 (Compulsory elective subject)
Identifier	3011365
Version	V2
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2023
Valid until	-
Module level	Bachelor
Content	Grundlagen der Wirtschafts- und Baubetriebslehre; Besonderheiten der Bauindustrie; Bedingungen der Bauproduktion; Die VOB; Organisationsstrukturen und Managementfunktionen; Baubetriebliches internes und externes Rechnungswesen; Kalkulation im Baubetrieb; Arbeitsvorbereitung, Angebots- und Auftragsmanagement; Projektabwicklung; Leistungsmeldung und Soll-Ist-Vergleich
Learning Objectives/ Learning Outcomes	Die Studierenden erlangen die Fähigkeit, komplexe Bauprojekte zu kalkulieren. Sie erwerben Kenntnisse über die Abwicklung von Bauprojekten. Die Studierenden erwerben Kenntnisse über die Betriebsabrechnung in Bauunternehmen. Die Studierenden erwerben Grundkenntnisse über die Durchführung von Wirtschaftlichkeitskontrollen bei Bauprojekten.
(Study-Specific) Prerequisites	-
(recommended) Requirements	-
References	Vorlesungsumdruck Realisierungsmanagement 2
Language	German
Examination Terms	Graded written exam. Admission requirements for participation in the written exam: passed term paper.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Katharina Klemt-Albert
ECTS Credits	3
Contact time (WSH)	2
Examination duration (min)	-
Total hours (h)	90,0
Contact hours (h)	30,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Construction Site Management 2 (301136501)	4th semester	no semester recommended	0	0
Written exam Construction Site Management 2 (301136502)	4th semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Tutorial Construction Site Management 2	4th semester	no semester recommended	-	2

Module titel	Railway Engineering (Compulsory elective subject)
Identifier	3020995
Version	V1
Duration (Semester)	two semesters
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2019
Valid until	-
Module level	Bachelor
Content	Railway Engineering I: Physical fundamentals of railways; track engineering; fundamentals of vehicle technology; driving dynamics and driving time calculation; routing elements; planning of railway lines; construction of the line in ground plan and longitudinal section; track construction methods; determination of earth masses. Railway Engineering II: Timetabling and operation management; elements of the stations and their design; switches and crossings; structuring of nodes, passenger and freight traffic facilities; train control systems; level crossings.
Learning Objectives/ Learning Outcomes	Railway Engineering I: Basic knowledge of railway track engineering; basic knowledge of track construction methods; basic knowledge of vehicle engineering and vehicle dynamics for railways; ability to plan and dimension components of railway track engineering; ability to route railways; ability to calculate earth masses. Railway Engineering II: Fundamentals of the design, layout and safety equipment of railway stations; fundamentals of the layout and safety of level crossings; fundamentals of operation management and timetabling; design of a track plan of small operating points, including the required main signals; calculation of switches in a track plan; dimensioning of track groups with the aid of a deterministic and a simple stochastic approach; fundamentals of the planning of railway networks.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdruck Eisenbahnwesen I; Vorlesungsumdruck Eisenbahnwesen II
Language	German
Examination Terms	Graded written exam. Prerequisite for the participation in the written exam is either the passed term paper on Railway Engineering II. If both papers are passed, a bonus can be applied to the first attempt at the exam. The exact criteria for the acquisition and awarding of the bonus will be announced at the beginning of the semester.
Miscellaneous	-
Module coordinator	Univ.-Prof. Nils Nießen
ECTS Credits	6
Contact time (WSH)	4
Examination duration (min)	-

Total hours (h)	180,0
Contact hours (h)	60,0
Self-study hours (h)	120,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Railway Engineering I (302099501)	4th semester	no semester recommended	0	0
Term paper Railway Engineering II (302099502)	5th semester	no semester recommended	0	0
Exam Railway Engineering (302099503)	5th semester	no semester recommended	6	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and exercise Railway Engineering I	4th semester	no semester recommended	-	2
Lecture and exercise Railway Engineering II	5th semester	no semester recommended	-	2

Module titel	Road Engineering and Constructional Engineering of Transport Infrastructure (Compulsory elective subject)
Identifier	3020999
Version	V2
Duration (Semester)	two semesters
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Straßenplanung I: Planungsmethodik; Entwurf von Straßen: Grundlagen der Fahrdynamik, Trassierung im Lageplan, Trassierung im Höhenplan, Sichtweiten (Halte-/Überholsichtweite), Grundlagen der Querschnittsgestaltung, Straßenentwässerung, Grundlagen der räumlichen Linienführung; Verkehrsflusstheorie: Grundlagen des Verkehrsablaufs, Berechnungsverfahren zum Verkehrsablauf; Bemessung von Straßenverkehrsanlagen: Grundlagen der verkehrstechnischen Bemessung, Bemessung von Autobahnabschnitten, Bemessung von Landstraßenabschnitten; Knotenpunktgestaltung: Grundlagen der Knotenpunktgestaltung, Bemessung von Knotenpunkten. Bautechnik von Verkehrsanlagen I: Entstehung, Bestandteile und Einteilung von Böden; Abbau, Transport und Einbau von Böden; Wasser und Frost, Grundprüfung des vorhandenen oder eingebauten Materials; Erdbauspezifische Belange von Verkehrswegen; Anforderung an den Untergrund/Unterbau, Bodenverbesserung/-verfestigung; Aufgabe, Funktion und Aufbau der Straßenkonstruktion sowie deren Dimensionierung; Mineralstoffe, künstliche Gesteine, Bautechnische Anforderungen an Gesteine, Güteüberwachung von Gesteinen; Bindemittel; Walzasphalt, Gussasphalt, Starre Befestigung (Be-ton); Bituminöse Prüfverfahren; Prüfung der fertigen Konstruktion und Anforderungen an die fertige Konstruktion; Asphalttechnologie.
Learning Objectives/ Learning Outcomes	Road Planning I: Independent design of roads; independent dimensioning of road traffic facilities taking into account traffic theory principles; basic understanding of the interrelationships in road and planning law; solution of planning law issues. Pavement Technology I: Independent dimensioning of road constructions; ability to select and design road construction materials; independent selection of testing concepts before, during and after realisation of road construction projects; insight into relevant guidelines/standards/regulations and their application.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Lecture notes Road Planning; Guidelines notes; Lecture notes on earthworks and road construction technology;
Language	German
Examination Terms	Graded exam „Road Engineering and Constructional Engineering of Transport Infrastructure“. Admission requirements for participation in the exam: passed preliminary exam requirements: „Road Planning I“ and „Constructional Engineering of Transport Infrastructure I“. The preliminary exam requirements can be completed alone or in combination with one of the following courses: term paper (6-8 problems), colloquium or e-test. The form of the preliminary examination requirements are to be announced at the beginning of the semester

– Transportation and Regional Planning

+ Road Engineering and Constructional Engineering of Transport ...

and should be completed during the regular semester. The preliminary exam requirements for the courses „Road Planning I“ and „Constructional Engineering of Transport Infrastructure I“ should be completed independently of each other in the regular semester.

Miscellaneous	-
Module coordinator	Prof. Dr. Alvaro Garcia-Hernandez
ECTS Credits	7
Contact time (WSH)	6
Examination duration (min)	-
Total hours (h)	210,0
Contact hours (h)	90,0
Self-study hours (h)	120,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Preliminary exam requirements Construction Technology of Traffic Arrangements I (302099901)	5th semester	no semester recommended	0	0
Preliminary exam requirements Road Planning I (302099902)	4th semester	no semester recommended	0	0
Exam Road Planning and Construction Technology of Traffic Arrangements (302099903)	5th semester	no semester recommended	7	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Tutorial Road Planning I	4th semester	no semester recommended	-	3
Lecture and Tutorial Construction Technology of Traffic Arrangements I	5th semester	no semester recommended	-	3

Module titel	Transportation and Traffic Planning I (Compulsory elective subject)
Identifier	3011367
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2009
Valid until	-
Module level	Bachelor
Content	This course teaches the basics of urban traffic planning. The focus is on the design of urban road spaces and the control of traffic at intersections. The lecture is interlinked with integrated exercise modules in which the sub-areas "road space design" and "urban traffic engineering" are deepened. In the exercises, the students have to record and evaluate a street section and prepare their own street design. In addition, a signal programme is independently designed for a traffic light-controlled junction and the traffic quality of the junction is then determined and the dimensioning of a small roundabout is carried out.
Learning Objectives/ Learning Outcomes	Students acquire knowledge of the basics of the design and operation of urban road traffic facilities and their links, as well as the relevant guidelines and regulations. They are enabled to create smaller road space designs and learn the theory and procedures for dimensioning urban junctions.
(Study-Specific) Prerequisites	None
(recommended) Requirements	Knowledge of 'Methodology of Planning' is recommended for participation.
References	Lecture notes Transport Planning I
Language	German
Examination Terms	Graded written exam. Admission requirements for participation in the written exam: ungraded passed term paper and colloquium.
Miscellaneous	-
Module coordinator	Dr.-Ing. Conny Louen
ECTS Credits	3
Contact time (WSH)	3
Examination duration (min)	60
Total hours (h)	90,0
Contact hours (h)	45,0
Self-study hours (h)	45,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Transportation and Traffic Planning I (301136701)	4th semester	no semester recommended	0	0
Written exam (or oral exam) Transportation and Traffic Planning I (301136702)	4th semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Tutorial Transportation and Traffic Planning I	4th semester	no semester recommended	-	3

Module titel	Transport Economics I (Compulsory elective subject)
Identifier	3011374
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2010
Valid until	-
Module level	Bachelor/Master
Content	Stakeholder on the transport market; transport statistics; introduction to transport modelling; transport market and mechanisms (passenger transport, freight transport); production scheduling in passenger transport; costs and revenues of transportation companies; market design and funding; principles of logistics and freight transport
Learning Objectives/ Learning Outcomes	The students learn the principles of transport economics and through that gain basic knowledge about how transportation companies and infrastructure companies operate. Furthermore, they should be able to predict the reactions of the transport market to changes in supply independently with the help of transport models.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdruck "Grundlagen der Verkehrswirtschaft"
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the exam.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Universitätsprofessor Dr.-Ing. Nils Nießen
ECTS Credits	2
Contact time (WSH)	2
Examination duration (min)	60
Total hours (h)	60,0
Contact hours (h)	30,0
Self-study hours (h)	30,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam Transport Economics I (301137401)	6th semester	no semester recommended	2	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Vorlesung und Übung Grundlagen der Verkehrswirtschaft	6th semester	no semester recommended	-	2

+ Introduction to CAD (3010748)

Module titel	Introduction to CAD (Compulsory elective subject)
Identifier	3010748
Version	V2
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2024
Valid until	-
Module level	Bachelor
Content	Fundamentals of CAD: creating, changing and deleting of basic elements (primitives) in 2D drawings; setup and use of complex element groups (cells) and their management in cell libraries; creation of surface elements; hatching and patterning of drawings; entity and use of reference drawings; dimensioning of linear and circular drawing objects; fundamentals of creating 3D drawings; working in three-dimensional coordinate systems; creation and manipulation of primitives in 3D drawings; reference drawings and cell libraries in conjunction with 3D designs; construction of B-spline curves and surfaces; creating of rotationally symmetric bodies; properties and use of auxiliary coordinate systems; derivation of intersection drawings from 3D models; visualization tools related to 3D designs; printing of technical drawings in predetermined scales
Learning Objectives/ Learning Outcomes	Basic understanding of computer aided drawing; assessing the advantages and disadvantages of CAD; ability to estimate the time effort; capability for autonomous creation of simple 2D and 3D drawings
(Study-Specific) Prerequisites	-
(recommended) Requirements	-
References	Anleitungsmanuskripte zu jedem Übungstermin
Language	German
Examination Terms	Graded electronic examination (drawing task on the computer). Prerequisites for participation in the examination: compulsory attendance of the exercises.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Universitätsprofessor Dr.-Ing. Jörg Blankenbach
ECTS Credits	3
Contact time (WSH)	2
Examination duration (min)	30
Total hours (h)	90,0
Contact hours (h)	30,0
Self-study hours (h)	60,0

+ Introduction to CAD (3010748)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Small group exercise Introduction to CAD (301074801)	2nd semester	no semester recommended	0	2
Exam Introduction to CAD (301074802)	2nd semester	no semester recommended	3	0

+ Data Analysis for Planning (3022398)

Module titel	Data Analysis for Planning (Compulsory elective subject)
Identifier	3022398
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2021
Valid until	-
Module level	Bachelor
Content	Darstellung von Daten; Statistische Tests (t-Test, Chi-Quadrat, Anova) und Testverteilungen; Konzeption von Erhebungen; Stichprobenumfänge; Aufbereitung von Daten aus Erhebungen und Befragungen; Gewichtung; Iterative Proportional Fitting; Regressions-verfahren (linear, nichtlinear, einfach, multipel); Discrete-Choice-Modellierung; Räumliche Regression; Zeitreihenanalyse; Survival Analysis; Faktorenanalyse; Clusteranalyse; Überblick über Big Data und Methoden der künstlichen Intelligenz
Learning Objectives/ Learning Outcomes	Überblick über statistische Verfahren zur Analyse von Daten für die Verkehrs- und Raumplanung mit Schwerpunkt auf sozio-ökonomischen und Verhaltensdaten; vertiefte Kenntnis zentraler Verfahren; Interpretation statistischer Analysen; verständliche und verwertbare Darstellung von Ergebnissen; Handhabung von Software zur Datenanalyse
(Study-Specific) Prerequisites	None
(recommended) Requirements	Recommended Knowledge: Applied Stastics.
References	-
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the exam.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Tobias Kuhnimhof
ECTS Credits	5
Contact time (WSH)	3
Examination duration (min)	-
Total hours (h)	150,0
Contact hours (h)	45,0
Self-study hours (h)	105,0

+ Data Analysis for Planning (3022398)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Data Analysis for Planning (302239801)	3rd semester	no semester recommended	5	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture Data Analysis for Planning	3rd semester	no semester recommended	-	2
Exercise Data Analysis for Planning	3rd semester	no semester recommended	-	1

+ Next generation fire control challenges and solutions (3017963)

Module titel	Next generation fire control challenges and solutions (Compulsory elective subject)
Identifier	3017963
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2018
Valid until	-
Module level	Bachelor
Content	Only by using the latest findings from science and technology can modern and sophisticated architecture using highly transparent building envelopes, complex building bodies with interior texture, and modern building materials be realized. This also applies to future-oriented fire protection. In this class, the phenomena and progression of fires are analyzed. Building upon that, the fundamentals of fire prevention, control and safety are taught, with the following focal points: structural and plant technical fire control as well as operational and organizational fire control. In cooperation with the teaching and research area Gender and Diversity in Engineering there will be a focus on topics such as plant fire control, fire alarm systems, extinguishing systems, and building services systems during the course. The class is a meaningful supplement and extension of other courses offered by the Faculty of Civil Engineering.
Learning Objectives/ Learning Outcomes	Students that take the course "Next Generation Fire Control Challenges and Solutions" are equipped with extensive knowledge and skills of a fire control engineer. They solve practical and complex problems of fire prevention in connection with all intersections of technical building equipment: They determine the principles of building laws for solving practical problems with appropriate risk implementation of technical regulations for planning and implementing projects while taking into account the aspects of preventative and organizational fire protection. They derive the complex requirements and challenges of the field of fire control in connection with all intersections of technical building equipment in context of construction projects. They assess and apply the methods, effects and general basics of technical fire control methods while taking into account the risks posed by fires and the specific factors as well as the interactions of these agents. They know fire control engineer methods and use these to achieve performance and protection oriented development of fire control solutions.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Landesbauordnung BauO NRW vom 15.12.2016. Sonderbauverordnung vom 02.12.2016. Muster-Industrie-Baurichtlinie vom Juli 2014. Mayr,J.; Battran, L.: Handbuch Brandschutzatlas, Grundlagen - Planung - Ausführung, Feuertrutz Verlag, 2.aktualisierte Auflage, Ausgabe: 2011(05). Schneider, Ulrich: Ingenieurmethoden im Baulichen Brandschutz, Expert-Verlag GmbH, 6. neu bearbeitete Auflage (21. März 2011). Brandschutzleitfaden des Bundes. Heidelberg, R.: Praxishandbuch Brandschutz im Bestand, Bewertung - Planung - Konzepte - Maßnahmen, Feuertrutz Verlag, 1.Auflage, Ausgabe 2012(11). BauPrüfVO in der jeweils aktuellen Fassung. AHO Heft 17
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the written exam.
Miscellaneous	-

+ Next generation fire control challenges and solutions (3017963)

Module coordinator	Modulverantwortlicher: Universitätsprofessorin Dr. phil. Carmen Leicht-Scholten
ECTS Credits	3
Contact time (WSH)	2
Examination duration (min)	90
Total hours (h)	90,0
Contact hours (h)	30,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Exam Next generation fire control challenges and solutions (301796301)	5th semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture Next generation fire control challenges and solutions	5th semester	no semester recommended	-	2

+ Introduction to Robotics (3026332)

Module titel	Introduction to Robotics (Compulsory elective subject)
Identifier	3026332
Version	-
Duration (Semester)	one semester
Cycle (Semester)	summer semester
Valid from	Summer semester 2022
Valid until	-
Module level	Bachelor
Content	1. History of Robots 2. Definition. What is a robot? 3. Kinematics Introduction 4. Main hardware parts: 4.1 Actuators and end-effectors. 4.2 Sensors. 4.3 Power (autonomy). 4.4 Computational power. 5. Main software parts: 5.1 Localisation. 5.2 Navigation. 5.3 Control. 6. Robotics in Simulation, including introduction to Simulation software. 7. Simulation practical: a robot rover navigating a cluttered environment, a robotic arm hammering task, a robotic arm bricklaying.
Learning Objectives/ Learning Outcomes	- Describe what a robot is. - List what are the main elements of a robot. - Identify different methods of software-related issues such as localisation, perception and control. - Ability to perform robotic tasks in a simulation programming environment.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Robotics. Modelling, Planning and Control Autoren: Siciliano, B., Sciavicco, L., Villani, L., Oriolo, G. 2009, Springer.

+ Introduction to Robotics (3026332)

Language	English
Examination Terms	The examination consists of a graded presentation (graded program-based elaboration consisting of several homework assignments and a graded presentation). The module grade is 75% based on the grade of the elaboration and 25% based on the grade of the presentation. There are no prerequisites for participation in the examination.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr. Raul Fuentes
ECTS Credits	3
Contact time (WSH)	2
Examination duration (min)	-
Total hours (h)	90,0
Contact hours (h)	30,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Presentation Introduction to Robotics (302633201)	summer semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Exercise Introduction to Robotics	summer semester	no semester recommended	-	2

+ Mechanics III (3010745)

Module titel	Mechanics III (Compulsory elective subject)
Identifier	3010745
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2012
Valid until	-
Module level	Bachelor
Content	Rheologische Modelle; Diskussion verschiedener Materialverhaltensweisen (z.B. Stahl, Beton, Asphalt) und Vereinfachung; Erarbeitung des Gleichungssatzes für eine eindimensionale Kraft-Verformungs-Analyse; Vorstellung verschiedener numerischer Methoden zur Gleichungslösung in einer Dimension; Vergleich Experiment - Simulation; Praktikum zum Selbstprogrammieren und -rechnen.
Learning Objectives/ Learning Outcomes	Grundlegendes Verständnis verschiedener Materialverhaltensweisen (elastoplastisch, viskoelastisch usw.); Übersicht über die Grundgleichungen der Mechanik (Kinematik, Gleichgewicht, Materialgesetz) in einer Dimension; Sicherheit in der Anwendung einfacher numerischer Methoden
(Study-Specific) Prerequisites	None
(recommended) Requirements	Knowledge of Mechanics I and Mechanics II is recommended for participation.
References	-
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the exam.
Miscellaneous	-
Module coordinator	Modulverantwortliche: Universitätsprofessorin Dr.-Ing. Stefanie Reese
ECTS Credits	3
Contact time (WSH)	2
Examination duration (min)	90
Total hours (h)	90,0
Contact hours (h)	30,0
Self-study hours (h)	60,0

+ Mechanics III (3010745)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam Mechanics III (301074501)	3rd semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Tutorial Mechanics III	3rd semester	no semester recommended	-	2

+ Physics (3010736)

Module titel	Physics (Compulsory elective subject)
Identifier	3010736
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2012
Valid until	-
Module level	Bachelor
Content	Units; force; motion; conservation of energy; oscillations and mechanical waves; temperature; heat and 1st law of thermodynamics; heat conduction, radiation and convection; ideal gases; entropy and 2nd law of thermodynamics
Learning Objectives/ Learning Outcomes	Basic knowledge of classical physics as a basis to understanding engineering issues. Using numerous tasks the elaboration of strategies of the principles is promoted up to a special solution.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Arbeitsunterlagen vom Lehrstuhl; Halliday Physik (ISBN 978-3-527-40645-6)
Language	German
Examination Terms	Graded written exam. There are no admission requirements for attending the exam.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Dr.-Ing. Dipl.-Phys. Joachim Hannawald
ECTS Credits	3
Contact time (WSH)	3
Examination duration (min)	90
Total hours (h)	90,0
Contact hours (h)	45,0
Self-study hours (h)	45,0

+ Physics (3010736)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Written exam: Physics (301073601)	1st semester	no semester recommended	3	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture: Physics	1st semester	no semester recommended	-	2
Tutorial: Physics	1st semester	no semester recommended	-	1

+ Introduction to Scientific Work in Engineering (3014059)

Module titel	Introduction to Scientific Work in Engineering (Compulsory elective subject)
Identifier	3014059
Version	V2
Duration (Semester)	one semester
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	In this course, the students are introduced to the basics of efficient scientific work: • Structure/composition of scientific papers • Citation methods • Literature research • Presentation techniques • Reading techniques • Learning techniques • Time management
Learning Objectives/ Learning Outcomes	Students should achieve the following learning outcomes: • Knowledge of methods, procedure and criteria of scientific work • Knowledge of correct citation • Understanding about structural and expedient reading, treatment and presentation • Successful, autonomous writing of a scientific work in the form of an ungraded term paper.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a ungraded term paper. Prerequisite for participation in the examination: Presence in the seminar is compulsory.
Miscellaneous	-
Module coordinator	Modulverantwortlicher: Universitätsprofessorin Dr. phil. Carmen Leicht-Scholten
ECTS Credits	3
Contact time (WSH)	2
Examination duration (min)	-
Total hours (h)	90,0
Contact hours (h)	30,0

+ Introduction to Scientific Work in Engineering (3014059)

Self-study hours (h) 60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term paper Introduction to Scientific Work in Engineering (301405901)	3rd semester	no semester recommended	3	0
Seminar Introduction to Scientific Work in Engineering (301405902)	3rd semester	no semester recommended	0	2

Module titel	Introduction to Finite Element Method for Structural Analysis (Compulsory elective subject)
Identifier	3019369
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2020
Valid until	Summer semester 2024
Module level	Bachelor
Content	• Energy methods and the principle of virtual displacements • Isoparametric concept • Variational formulation of finite elements • Finite elements for trusses and beams • Locking and alternative FE formulations • Linear equation solvers
Learning Objectives/ Learning Outcomes	At the end of the course, you can • understand the structure of a FE program. • create your own FE program. • code your own functions in Matlab. • use energy methods and the principle of virtual displacements. • transform differential equations into the weak form. • develop your own finite elements. • understand the cause of locking-effects. • modify element formulations to reduce locking effects. • understand the way solvers for linear equation systems work and know their limitations.
(Study-Specific) Prerequisites	None
(recommended) Requirements	-
References	Vorlesungsumdrucke
Language	German
Examination Terms	The examination consists of a graded written exam and a graded term paper. The grade is made up of 80% of the written exam and 20% of the paper. Prerequisite for participation in the written exam: passed term paper.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. habil. Sven Klinkel
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	120,0

- Construction Project Life Cycle: Design ...
+ Introduction to Finite Element Method for Structural Analysis ...

Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term Paper Introduction to Finite Element Method for Structural Analysis (301936901)	no semester recommended	no semester recommended	0	0
Exam Introduction to Finite Element Method for Structural Analysis (301936902)	no semester recommended	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Exercise Introduction to Finite Element Method for Structural Analysis	5th semester	no semester recommended	-	4

Module titel	Introduction to Finite Element Method for Structural Analysis (Compulsory elective subject)
Identifier	3019369
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2023
Valid until	-
Module level	Bachelor
Content	• Energy methods and the principle of virtual displacements • Isoparametric concept • Variational formulation of finite elements • Finite elements for trusses and beams • Locking and alternative FE formulations • Linear equation solvers
Learning Objectives/ Learning Outcomes	At the end of the course, you can • understand the structure of a FE program. • create your own FE program. • code your own functions in Matlab. • use energy methods and the principle of virtual displacements. • transform differential equations into the weak form. • develop your own finite elements. • understand the cause of locking-effects. • modify element formulations to reduce locking effects. • understand the way solvers for linear equation systems work and know their limitations.
(Study-Specific) Prerequisites	-
(recommended) Requirements	-
References	Vorlesungsumdrucke
Language	German
Examination Terms	The examination consists of a graded written exam and a graded term paper. The grade is made up of 80% of the written exam and 20% of the paper. Prerequisite for participation in the written exam: passed term paper.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. habil. Sven Klinkel
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	120,0

— Integral Planning or Introduction to ...
+ Introduction to Finite Element Method for Structural Analysis ...

Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Term Paper Introduction to Finite Element Method for Structural Analysis (301936901)	5th semester	no semester recommended	0	0
Exam Introduction to Finite Element Method for Structural Analysis (301936902)	5th semester	no semester recommended	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Exercise Introduction to Finite Element Method for Structural Analysis	5th semester	no semester recommended	-	4

Module titel	Integral Planning (Compulsory elective subject)
Identifier	3028324
Version	V1
Duration (Semester)	one semester
Cycle (Semester)	winter semester
Valid from	Winter semester 2023
Valid until	-
Module level	Bachelor/Master
Content	The course is designed to provide students with basic knowledge and methodological skills for the planning of building structures. The focus is on the early service phases in the project. The focus is on the necessary interaction and communication with the project participants and the impact of planning decisions on the aspects of functionality, costs and sustainability. The contents of the module are divided into 2 core areas: General methodological competencies and exemplary implementation through business game / presentation. Within the lecture, students are taught the following theoretical basics: • General: From the idea to the description of requirements - Sequence and early phases of a building construction project and its interfaces. • Introduction to the preparation of utilization concept and requirements specification • Introduction to various digital and parameterizable planning methods and their possible applications in the project • Implications and subject of integral planning / early exchange of all involved sub-trades as an essential part of integral planning • Leadership of interdisciplinary knowledge transfer, reduction of conflict potential • Communication and cooperation - various aspects of professional exchange in project work • Complexity reduction in communication and project organization in cooperation The acquired basics are applied in a business game or group work. In doing so, the students deal with the following thematic focal points in an exemplary manner: • Collaborative solution of exemplary tasks/problems of planning in early performance phases. • Prerequisites, components, interfaces (information technology) • Creation of structural concept and material selection • Location and dimension of technical centers on the basis of an energy concept • Exemplary route planning / coordination with other trades • Iterative coordination with other technical planners • Dealing with planning changes • Contract design with BIM, effects on conventional contract models and project examples
Learning Objectives/ Learning Outcomes	After completing the module, students should be able to: • understand what integral planning means in the construction industry and what role it plays, especially in early service phases • select working methods appropriate to the project requirements • understand the interdependencies between the different trades • recognize and implement the need for communication and coordination between trades • be able to communicate complex, trade-specific content to other project participants • Identify project requirements from the client's point of view and derive planning requirements from them, taking into account sustainability and cost aspects. • know methods/tools for integral planning in early phases of work and to be able to apply them exemplarily.

— Integral Planning or Introduction to ...
+ Integral Planning (3028324)

	- to develop an understanding of methods, constraints/conflicts and interfaces between trades - put oneself in other roles and to be able to work in a team as well as independently/ integral planning
(Study-Specific) Prerequisites	-
(recommended) Requirements	Recommended knowledge: Project &; Contract Management, Structural Engineering, Introduction to Civil Engineering
References	Documents from lectures and exercises
Language	German
Examination Terms	The examination consists of a graded presentation ("Referat"). The grade is composed of 50% written work and 50% oral presentation. There are no prerequisites for taking the exam.
Miscellaneous	-
Module coordinator	Univ.-Prof. Dr.-Ing. Katharina Klemm-Albert
ECTS Credits	4
Contact time (WSH)	4
Examination duration (min)	-
Total hours (h)	120,0
Contact hours (h)	60,0
Self-study hours (h)	60,0

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Seminar Business Simulation Integral Planning (302832401)	5th semester	5th semester	0	0
Exam Integral Planning (302832402)	5th semester	5th semester	4	0

▲ Offer node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Lecture and Exercise Integral Planning	5th semester	5th semester	-	4

- Relevant Additional Subjects for Studies ...
+ Relevant Additional Subjects for Studies Abroad (3022699)

Module titel	Relevant Additional Subjects for Studies Abroad (Elective subject)
Identifier	3022699
Version	V1
Duration (Semester)	-
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Je nach Fächerwahl Dieses Modul ermöglicht die Anrechnung von bis zu 8 ECTS Kreditpunkten aus einem studienbezogenen Auslandsaufenthalt, sofern die im Ausland absolvierten Veranstaltungen die folgenden Kriterien erfüllen: Die Lernziele und Inhalte der im Ausland absolvierten Prüfungsleistungen haben einen direkten Bezug zu den studiengangspezifischen Studienzielen des Studiengangs an der Fakultät für Bauingenieurwesen. Die Lernziele und Inhalte der im Ausland absolvierten Veranstaltungen stellen eine inhaltliche Ergänzung zum Curriculum des Studiengangs der Fakultät für Bauingenieurwesen dar, gehen also über das curricular verankerte Veranstaltungsangebot hinaus.
Learning Objectives/ Learning Outcomes	Die Studierenden erwerben interkulturelle Kompetenzen und vertiefen ihre fremdsprachlichen Kenntnisse. Durch die internationale Fachperspektive auf ihr Studium und die Wissenschaftswelt sind sie in der Lage, ihre zukünftige berufliche Tätigkeit in einem globalen Kontext zu betrachten.
(Study-Specific) Prerequisites	-
(recommended) Requirements	Die Anerkennung innerhalb dieses Moduls sollte in der Regel vor Antritt des Auslandsaufenthalts vom zuständigen Prüfungsausschuss zugestimmt werden. Dazu werden die im Ausland geplanten Prüfungsleistungen und deren Anrechnung in einem Learning Agreement zwischen den drei beteiligten Parteien (Studierende bzw. Studierender, Gasthochschule und Heimathochschule) festgehalten. Die Studierenden sollten beim zuständigen Prüfungsausschuss die für den Abschluss des Learning Agreements erforderlichen Angaben zu den im Ausland geplanten Kursen rechtzeitig machen.
References	Entsprechend den Empfehlungen in den gewählten Modulen oder Kursen.
Language	-
Examination Terms	Prüfungsbedingungen entsprechend den geforderten Prüfungsbedingungen der im Ausland gewählten Module oder Kurse.
Miscellaneous	-
Module coordinator	-
ECTS Credits	-
Contact time (WSH)	-

— Relevant Additional Subjects for Studies ...

+ Relevant Additional Subjects for Studies Abroad (3022699)

Examination duration (min)	-
Total hours (h)	-
Contact hours (h)	-
Self-study hours (h)	-

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269901)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269902)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269903)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269904)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269905)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269906)	no semester recommended	no semester recommended	-	-

- Relevant Additional Subjects for Studies ...
+ Relevant Additional Subjects for Studies Abroad (3022699)

Module titel	Relevant Additional Subjects for Studies Abroad (Elective subject)
Identifier	3022699
Version	V1
Duration (Semester)	-
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2020
Valid until	-
Module level	Bachelor
Content	Je nach Fächerwahl Dieses Modul ermöglicht die Anrechnung von bis zu 8 ECTS Kreditpunkten aus einem studienbezogenen Auslandsaufenthalt, sofern die im Ausland absolvierten Veranstaltungen die folgenden Kriterien erfüllen: Die Lernziele und Inhalte der im Ausland absolvierten Prüfungsleistungen haben einen direkten Bezug zu den studiengangspezifischen Studienzielen des Studiengangs an der Fakultät für Bauingenieurwesen. Die Lernziele und Inhalte der im Ausland absolvierten Veranstaltungen stellen eine inhaltliche Ergänzung zum Curriculum des Studiengangs der Fakultät für Bauingenieurwesen dar, gehen also über das curricular verankerte Veranstaltungsangebot hinaus.
Learning Objectives/ Learning Outcomes	Die Studierenden erwerben interkulturelle Kompetenzen und vertiefen ihre fremdsprachlichen Kenntnisse. Durch die internationale Fachperspektive auf ihr Studium und die Wissenschaftswelt sind sie in der Lage, ihre zukünftige berufliche Tätigkeit in einem globalen Kontext zu betrachten.
(Study-Specific) Prerequisites	-
(recommended) Requirements	Die Anerkennung innerhalb dieses Moduls sollte in der Regel vor Antritt des Auslandsaufenthalts vom zuständigen Prüfungsausschuss zugestimmt werden. Dazu werden die im Ausland geplanten Prüfungsleistungen und deren Anrechnung in einem Learning Agreement zwischen den drei beteiligten Parteien (Studierende bzw. Studierender, Gasthochschule und Heimathochschule) festgehalten. Die Studierenden sollten beim zuständigen Prüfungsausschuss die für den Abschluss des Learning Agreements erforderlichen Angaben zu den im Ausland geplanten Kursen rechtzeitig machen.
References	Entsprechend den Empfehlungen in den gewählten Modulen oder Kursen.
Language	-
Examination Terms	Prüfungsbedingungen entsprechend den geforderten Prüfungsbedingungen der im Ausland gewählten Module oder Kurse.
Miscellaneous	-
Module coordinator	-
ECTS Credits	-
Contact time (WSH)	-

— Relevant Additional Subjects for Studies ...

+ Relevant Additional Subjects for Studies Abroad (3022699)

Examination duration (min)	-
Total hours (h)	-
Contact hours (h)	-
Self-study hours (h)	-

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269901)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269902)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269903)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269904)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269905)	no semester recommended	no semester recommended	-	-
Sinnvolle fachliche Ergänzung aus studienbezogenen Auslandsaufenthalten (302269906)	no semester recommended	no semester recommended	-	-

+ Bachelor Thesis (3013481)

Module titel	Bachelor Thesis (Compulsory subject)
Identifier	3013481
Version	Angelegt über RWTH API als 1
Duration (Semester)	one semester
Cycle (Semester)	winter/summer semester
Valid from	Winter semester 2012
Valid until	-
Module level	Bachelor
Content	Die Bachelor-Arbeit ist eine schriftliche Arbeit der Kandidatin bzw. des Kandidaten. Sie soll zeigen, dass die Kandidatin bzw. der Kandidat in der Lage ist, ein Problem aus dem Bereich des Bauingenieurwesens innerhalb einer vorgegebenen Frist nach wissenschaftlichen Methoden unter Anleitung selbstständig zu bearbeiten.
Learning Objectives/ Learning Outcomes	-
(Study-Specific) Prerequisites	Prerequisite for participation in the examination are the 125 credits to be achieved, the passed modules "Mathematics I, Mathematics II, Mechanics I and Mechanics II" and passed and by the internship office recognized practical work.
(recommended) Requirements	-
References	-
Language	German
Examination Terms	The examination consists of a graded written paper with a lecture colloquium.
Miscellaneous	-
Module coordinator	-
ECTS Credits	12
Contact time (WSH)	0
Examination duration (min)	0
Total hours (h)	360,0
Contact hours (h)	,0
Self-study hours (h)	360,0

+ Bachelor Thesis (3013481)

● Exam node

Title	Recommended Semester (Study start winter)	Recommended Semester (Study start summer)	ECTS Credits	Contact time (WSH)
Bachelor Thesis (301348101)	6th semester	no semester recommended	12	0